

# MTH 9875 The Volatility Surface: Fall 2026

## Lecture 3: Affine models

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## Outline of lecture

- Affine forward variance models
- The classical Heston model
- The rough Heston model
  - Non-Markovianity of the rough Heston model
- Laplace transforms and the resolvent identity
- The characteristic function of an affine forward variance model
  - Classical Heston and rough Heston characteristic functions
- Implementation of the classical and rough Heston models in R
- Numerical experiments

## The classical Heston model

Once again, the classical Heston stochastic volatility model may be written as

$$\begin{aligned}\frac{dS_t}{S_t} &= \sqrt{V_t} dZ_t \\ dV_t &= -\lambda (V_t - \bar{V}) dt + \eta \sqrt{V_t} dW_t\end{aligned}$$

with  $d\langle Z, W \rangle_t = \rho dt$  and where  $\lambda$  is the speed of reversion of  $V_t$  to its long term mean  $\bar{V}$ .

- The process followed by the instantaneous variance  $V_t$  may be recognized as a version of the square root process or CIR process of [Cox, Ingersoll, Ross]<sup>[6]</sup>.
- It is a (jump-free) special case of a so-called *affine jump diffusion* (AJD).

- Roughly speaking a jump-diffusion process for which the drifts and covariances and jump intensities are linear in the state vector (which is  $\{S, V\}$  in this case).

## Forward variance in the Heston model

With  $\xi_t(u) = \mathbb{E}_t[V_u]$ , take conditional expectations in the Heston variance SDE

$$dV_t = -\lambda(V_t - \bar{V}) dt + \eta \sqrt{V_t} dW_t.$$

to get

$$d_u \xi_t(u) = -\lambda(\xi_t(u) - \bar{V}) du.$$

This ODE has the solution

$$\xi_t(u) = (\xi_t(t) - \bar{V}) e^{-\lambda(u-t)} + \bar{V} = (V_t - \bar{V}) e^{-\lambda(u-t)} + \bar{V}.$$

## The Heston model in forward variance form

For each  $u$ ,  $\xi_t(u)$  is a conditional expectation and so a martingale in  $t$ . It is then immediate from the last equation that

$$d_t \xi_t(u) = \eta e^{-\lambda(u-t)} \sqrt{V_t} dW_t.$$

- It is easy to check explicitly that all drift (i.e.  $dt$ ) terms cancel.

## The rough Heston model

By considering the limit of a simple Hawkes process-based model of order flow, [El Euch and Rosenbaum]<sup>[7]</sup> derive a rough Heston model. The equation for variance in this model takes the form

$$V_u = \theta_t(u) - \frac{\lambda}{\Gamma(\alpha)} \int_t^u (u-s)^{\alpha-1} V_s ds + \frac{\nu}{\Gamma(\alpha)} \int_t^u (u-s)^{\alpha-1} \sqrt{V_s} dW_s.$$

where  $\alpha = H + \frac{1}{2}$ .

- $H \in (0, \frac{1}{2}]$  is the Hurst exponent of the volatility,  $\lambda > 0$  is the mean reversion parameter,  $\nu > 0$  is the volatility of volatility parameter.
- The function  $\theta$  is assumed to be continuous and represents a time-dependent mean reversion level.
- The rough Heston model generalizes the classical Heston model which is recovered when  $H = 1/2$ .

## Forward variance in the rough Heston model

- We first consider the special case  $\lambda = 0$ . In this case,  $\xi_t(u) = \mathbb{E}_t[V_u] = \theta_t(u)$ .
- It follows that

$$V_u = \xi_t(u) + \frac{\nu}{\Gamma(\alpha)} \int_t^u (u-s)^{\alpha-1} \sqrt{V_s} dW_s.$$

- Also

$$V_u = \xi_{t+h}(u) + \frac{\nu}{\Gamma(\alpha)} \int_{t+h}^u (u-s)^{\alpha-1} \sqrt{V_s} dW_s.$$

## The rough Heston model in forward variance form

Subtracting these two equations gives

$$\xi_{t+h}(u) - \xi_t(u) = \frac{\nu}{\Gamma(\alpha)} \int_t^{t+h} (u-s)^{\alpha-1} \sqrt{V_s} dW_s.$$

Taking the limit  $h \rightarrow 0$ , we obtain

$$d_t \xi_t(u) = \frac{\nu}{\Gamma(\alpha)} (u-t)^{\alpha-1} \sqrt{V_t} dW_t,$$

the rough Heston model in forward variance form.

## Non-Markovianity of the rough Heston model

- Note that the limit  $u \rightarrow t$  of the rough Heston model makes no sense.
  - This reflects the fact that the rough Heston model is not Markovian.
    - There is no SDE for  $V_t$  and no corresponding PDE.
  - On the other hand, we can write an SDE for each  $\xi_t(u)$ ,  $u > t$ .
    - We can even apply Itô's Lemma!
- The rough Heston model is Markovian in the infinite-dimensional forward variance curve  $\xi_t(u)$ ,  $u > t$ .

## Markovian, non-Markovian, and path-dependence

In forward variance form, the Heston model reads

$$d\xi_t(u) = \eta e^{-\lambda(u-t)} \sqrt{V_t} dW_t.$$

Thus

$$\begin{aligned}
 V_u &= \bar{V} + \int_{-\infty}^u \eta e^{-\lambda(u-s)} \sqrt{V_s} dW_s \\
 &= \bar{V} + \int_{-\infty}^t \eta e^{-\lambda(u-s)} \sqrt{V_s} dW_s + \int_t^u \eta e^{-\lambda(u-s)} \sqrt{V_s} dW_s \\
 &= \bar{V} + e^{-\lambda(u-t)} (V_t - \bar{V}) + \int_t^u \eta e^{-\lambda(u-s)} \sqrt{V_s} dW_s.
 \end{aligned}$$

- We see explicitly that  $V$  is Markovian. In other words, that  $\mathbb{E}[V_u | \mathcal{F}_t] = \mathbb{E}[V_u | V_t]$ .

Now consider a model with a different kernel, namely

$$\begin{aligned}
 d\xi_t(u) &= \left\{ \eta_1 e^{-\lambda_1(u-t)} + \eta_2 e^{-\lambda_2(u-t)} \right\} \sqrt{V_t} dW_t \\
 &=: \kappa(u-t) \sqrt{V_t} dW_t.
 \end{aligned}$$

Repeating the above analysis gives

$$\begin{aligned}
 V_u &= \bar{V} + \int_{-\infty}^t \left\{ \eta_1 e^{-\lambda_1(u-s)} + \eta_2 e^{-\lambda_2(u-s)} \right\} \sqrt{V_s} dW_s \\
 &\quad + \int_t^u \kappa(u-s) \sqrt{V_s} dW_s \\
 &= \bar{V} + \eta_1 e^{-\lambda_1(u-t)} \int_{-\infty}^t e^{-\lambda_1(t-s)} \sqrt{V_s} dW_s \\
 &\quad + \eta_2 e^{-\lambda_2(u-t)} \int_{-\infty}^t e^{-\lambda_2(t-s)} \sqrt{V_s} dW_s \\
 &\quad + \int_t^u \kappa(u-s) \sqrt{V_s} dW_s.
 \end{aligned}$$

But in this case,

$$V_t - \bar{V} = \int_{-\infty}^t \left\{ \eta_1 e^{-\lambda_1(u-t)} + \eta_2 e^{-\lambda_2(u-t)} \right\} \sqrt{V_s} dW_s.$$

Substituting back gives

$$\begin{aligned}
 V_u &= \bar{V} + e^{-\lambda_1(u-t)} (V_t - \bar{V}) \\
 &\quad + \eta_2 \left( e^{-\lambda_2(u-t)} - e^{-\lambda_1(u-t)} \right) \int_{-\infty}^t e^{-\lambda_2(t-s)} \sqrt{V_s} dW_s \\
 &\quad + \int_t^u \kappa(u-s) \sqrt{V_s} dW_s.
 \end{aligned}$$

- $V_u$  depends on more than just  $V_t$  now. We need extra information about the historical path of  $W$ .
- We can show that  $\mathbb{E}_t[V_u]$  depends not only on  $V_t$  but on an exponentially weighted average of  $V_s$ ,  $s < t$ .
- So  $V$  is strictly speaking no longer Markov, though it is if we extend the state-space to include the weighted average.

## The power-law kernel

- We represent the power law kernel in the form (recall that  $\alpha = H + \frac{1}{2}$ )

$$\frac{\eta \tau^{H-1/2}}{\Gamma(H+1/2)} = \frac{\eta}{\Gamma(H+1/2) \Gamma(1/2-H)} \int_0^\infty \lambda^{-(H+1/2)} e^{-\lambda \tau} d\lambda$$

$$\approx \sum_i \eta_i e^{-\lambda_i \tau}.$$

- A very crude approximation to this kernel could be given by the sum of two exponentials we just studied.
- What this means is that a rough Heston model can be approximated as a sum of classical Heston models, all driven by the same Brownian motion.
  - [Abi Jaber and el Euch]<sup>[1]</sup> showed how to choose the coefficients  $\eta_i$  and  $\lambda_i$  so as to get good convergence of the sum of classical Heston models to the rough Heston model.

## Affine processes

The following explanation is due to Martin Keller-Ressel:

An *affine process* can be described as a Markov process whose log-characteristic function is an affine function of its initial state vector.

And here's a definition of the word *affine* from Wikipedia:

In geometry, an affine transformation or affine map or an affinity (from the Latin, *affinis*, "connected with") between two vector spaces (strictly speaking, two affine spaces) consists of a linear transformation followed by a translation:

$$x \mapsto Ax + b$$

## Affine CGF

Let  $X_t = \log S_t$ . According to Definition 2.2 of [Gatheral and Keller-Ressel]<sup>[10]</sup>, we say that a forward variance model has an *affine cumulant generating function* determined by  $g(t; u)$ , if its conditional cumulant generating function is of the form

(1)

$$\log \mathbb{E}_t \left[ e^{u(X_T - X_t)} \right] = \int_t^T g(T - s; u) \xi_t(s) ds.$$

for all  $u \in [0, 1]$ ,  $0 \leq t \leq T$  and  $g(\cdot; u)$  is  $\mathbb{R}_{\leq 0}$ -valued and continuous on  $[0, T]$  for all  $T > 0$  and  $u \in [0, 1]$ .

- The restriction  $u \in [0, 1]$  is for mathematical convenience.

## When is a forward variance model affine?

Theorem 2.4 of [Gatheral and Keller-Ressel]<sup>[9]</sup> states that a forward variance model has an affine CGF if and only if it takes the form

$$\begin{aligned} \frac{dS_t}{S_t} &= \sqrt{V_t} dZ_t \\ d\xi_t(u) &= \kappa(u - t) \sqrt{V_t} dW_t \end{aligned}$$

for some deterministic, non-negative decreasing kernel  $\kappa$ , which satisfies  $\int_0^T \kappa(r) dr < \infty$  for all  $T > 0$ .

Moreover,  $g(\cdot, u) : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\leq 0}$  in the definition (1) of the CGF is the unique global continuous solution of the convolution Riccati equation

$$g(\tau; u) = \frac{1}{2}(u^2 - u) + \rho u (\kappa \star g)(\tau; u) + \frac{1}{2} (\kappa \star g)(\tau; u)^2, \quad \tau \geq 0$$

where the convolution is given by

$$(\kappa \star g)(t; u) = \int_0^t \kappa(t - s) g(s; u) ds.$$

## Derivation of the Riccati equation

From the definition (1) of the CGF,

$$M_t = \mathbb{E}_t \left[ e^{u X_T} \right] = \exp \left\{ u X_t + \int_t^T \xi_t(s) g(T - s; u) ds \right\} =: \exp \{ u X_t + G_t \}$$

is a conditional expectation and thus a martingale in  $t$ .

Applying Itô's Lemma to  $M$  gives

$$\frac{dM_t}{M_t} = u dX_t + dG_t + \frac{u^2}{2} d\langle X \rangle_t + \frac{1}{2} d\langle G \rangle_t + u d\langle X, G \rangle_t.$$

Now

$$\begin{aligned} dX_t &= -\frac{1}{2} V_t dt + \sqrt{V_t} dZ_t \\ dG_t &= -\xi_t(t) g(T-t; u) dt + \int_t^T d\xi_t(s) g(T-s; u) ds \\ &= -V_t g(T-t; u) dt + \int_t^T \kappa(s-t) \sqrt{V_t} dW_t g(T-s; u) ds. \end{aligned}$$

We compute

$$\begin{aligned} d\langle X \rangle_t &= V_t dt \\ d\langle G \rangle_t &= V_t dt \left( \int_t^T \kappa(s-t) g(T-s; u) ds \right)^2 \\ d\langle X, G \rangle_t &= \rho V_t dt \int_t^T \kappa(s-t) g(T-s; u) ds. \end{aligned}$$

Imposing  $\mathbb{E}[dM_t] = 0$  and letting  $\tau = T - t$  gives

$$0 = V_t dt \left\{ -\frac{1}{2} u + \frac{1}{2} u^2 - g(\tau; u) + \rho u (\kappa \star g)(\tau; u) + \frac{1}{2} (\kappa \star g)(\tau; u)^2 \right\}$$

where the convolution integral is given by

$$(\kappa \star g)(\tau; u) = \int_0^\tau \kappa(\tau-s) g(s; u) ds.$$

- It is almost obvious why the CGF is affine if and only if the forward variance process is of the form  $d\xi_t(u) = \sqrt{V_t} \kappa(u-t) dW_t$ .

## The convolution Riccati equation

Rearranging gives

$$g(\tau; u) = \frac{1}{2} u(u-1) + \rho u (\kappa \star g)(\tau; u) + \frac{1}{2} (\kappa \star g)(\tau; u)^2,$$

as required.

## An aside: Fractional calculus

Define the fractional integral and differential operators:

$$I^\alpha f(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} f(s) ds; \quad D^\alpha f(t) = \frac{d}{dt} I^{1-\alpha} f(t).$$

The fractional integral is a natural generalization of the ordinary integral using the Cauchy formula for repeated integration:

$$\begin{aligned} I^n f(t) &:= \int_0^t dt_1 \int_0^{t_1} \dots dt_{n-1} \int_0^{t_{n-1}} f(t_n) dt_n \\ &= \frac{1}{(n-1)!} \int_0^t (t-s)^{n-1} f(s) ds. \end{aligned}$$

The induction step

$$\begin{aligned} I^{n+1} f(t) &:= \int_0^t I^n f(s) ds \\ &= \frac{1}{(n-1)!} \int_0^t ds \int_0^s (s-r)^{n-1} f(r) dr \\ &= \frac{1}{(n-1)!} \int_0^t f(r) dr \int_r^t (s-r)^{n-1} ds \\ &= \frac{1}{n!} \int_0^t (t-r)^n f(r) dr. \end{aligned}$$

### Example: The rough Heston model (with $\lambda = 0$ )

In this case, with  $\alpha = H + \frac{1}{2}$ ,  $\kappa(\tau) = \frac{\nu}{\Gamma(\alpha)} \tau^{\alpha-1}$  (the *power-law* kernel) and

$$\begin{aligned} \eta h(\tau; u) &:= (\kappa \star g)(\tau; u) = \frac{\nu}{\Gamma(\alpha)} \int_0^\tau (\tau-s)^{\alpha-1} g(s; u) ds \\ &= \nu I^\alpha g(\tau; u). \end{aligned}$$

Inverting this gives  $g(\tau; u) = D^\alpha h(\tau; u)$ .

The convolution integral Riccati equation then reads

$$D^\alpha h(\tau; u) = \frac{1}{2} u(u-1) + \rho \nu u h(\tau; u) + \frac{1}{2} \nu^2 h(\tau; u)^2,$$

consistent with [El Euch and Rosenbaum]<sup>[6]</sup>.

### The Mittag-Leffler kernel

The Mittag-Leffler kernel is given by



$$\kappa(\tau) = \nu \tau^{\alpha-1} E_{\alpha,\alpha}(-\lambda \tau^\alpha),$$

where  $E_{\alpha,\beta}$  is the generalized Mittag-Leffler function

$$E_{\alpha,\beta}(z) = \sum_{k \geq 0} \frac{z^k}{\Gamma(\alpha k + \beta)}.$$

- Although it looks complicated, the Mittag-Leffler function is a simple fractional generalization of the exponential function.
  - When  $\alpha = 1$  and  $\beta = 1$ ,  $\Gamma(\alpha k + \beta) = n!$ .

## Two limiting cases

Two special cases of  $\kappa(\tau) = \nu \tau^{\alpha-1} E_{\alpha,\alpha}(-\lambda \tau^\alpha)$ :

- $\lambda = 0$ :  $\kappa(\tau) = \frac{\nu}{\Gamma(\alpha)} \tau^{\alpha-1}$ 
  - the power-law kernel.
- $\alpha = 1$ :  $\kappa(\tau) = \nu e^{-\lambda \tau}$ 
  - the exponential kernel of classical Heston.

## Laplace transforms

Write  $\kappa_0$  for the power-law kernel and  $\kappa$  for the Mittag-Leffler kernel. Their Laplace transforms are

$$\hat{\kappa}_0(p) = \frac{\nu}{p^\alpha}, \quad \hat{\kappa}(p) = \frac{\nu}{p^\alpha + \lambda}.$$

- The second is the defining property of  $E_{\alpha,\alpha}$ : it is *the* function whose Laplace transform is  $1/(p^\alpha + \lambda)$ .
- Now with  $\lambda' = \lambda/\nu$ ,

$$\frac{1}{\kappa} - \frac{1}{\kappa_0} = \lambda' = \frac{\kappa_0 - \kappa}{\kappa \kappa_0},$$

so

$$\kappa = \frac{\kappa_0}{1 + \lambda' \kappa_0}.$$

## The rough Heston model (with $\lambda > 0$ )

Consider again the integral equation that emerges in the continuous time limit of the Hawkes microstructure model:

$$V_u = \theta_t(u) - \frac{\lambda}{\Gamma(\alpha)} \int_t^u (u-s)^{\alpha-1} V_s ds + \frac{\nu}{\Gamma(\alpha)} \int_t^u (u-s)^{\alpha-1} \sqrt{V_s} dW_s.$$

- $V$  is on both sides of this equation!

We can write the equation in convolution notation as:

$$V_u = \theta_t(u) - \lambda' (\kappa_0 \star V)_u + (\kappa_0 \star \sqrt{V} dW)_u.$$

Taking Laplace transforms:

$$\hat{V} = \hat{\theta} - \lambda' \hat{\kappa}_0 \hat{V} + \hat{\kappa}_0 \widehat{\sqrt{V} dW}.$$

Solving for  $\hat{V}$ , we get

$$\hat{V} = \hat{\theta} - \lambda' \frac{\hat{\kappa}_0}{1 + \lambda' \hat{\kappa}_0} \hat{\theta} + \frac{\hat{\kappa}_0}{1 + \lambda' \hat{\kappa}_0} \widehat{\sqrt{V} dW} \quad (1)$$

$$= \hat{\theta} - \lambda' \hat{\kappa} \hat{\theta} + \hat{\kappa} \widehat{\sqrt{V} dW}. \quad (2)$$

Inverting now gives

$$V_u = \theta_t(u) - \lambda' \int_t^u \kappa(u-s) \theta_s ds + \int_t^u \kappa(u-s) \sqrt{V_s} dW_s,$$

where  $\kappa$  is the Mittag-Leffler kernel.

## The resolvent identity

We just used the resolvent identity:

### The resolvent identity

$$\kappa_0 - \kappa = \lambda' \kappa_0 \star \kappa, \quad \lambda' = \lambda/\nu.$$

- The Mittag-Leffler kernel is the **resolvent** of the power-law kernel.

## The rough Heston model in forward variance form

We may write

$$V_u = \int_t^u \kappa(u-s) \sqrt{V_s} dW_s + \text{BV},$$

where BV represents terms of bounded variation.

Then

$$d\xi_t(u) = d\mathbb{E}_t[V_u] = \kappa(u-t) \sqrt{V_t} dW_t.$$

- Adding the drift  $\lambda$  has the effect of replacing the power-law kernel  $\kappa_0$  with the Mittag-Leffler kernel  $\kappa$ .

## The rough Heston fractional Riccati equation (with $\lambda > 0$ ).

As in the power-law  $\lambda = 0$  case, let  $\nu h(\tau; u) := (\kappa \star g)(\tau; u)$ . Recall that  $I^\alpha f = \frac{1}{\nu} \kappa_0 \star f$ .

Using the resolvent identity and associativity of convolution,

$$\begin{aligned} h &= \frac{1}{\nu} (\kappa \star g) = \frac{1}{\nu} (\kappa_0 \star g - \lambda' \kappa_0 \star \kappa \star g) \\ &= \frac{1}{\nu} \kappa_0 \star g - \frac{\lambda'}{\nu} \kappa_0 \star (\nu h) \\ &= I^\alpha g - \lambda I^\alpha h. \end{aligned}$$

Inverting gives

$$D^\alpha h(\tau; u) = g(\tau; u) - \lambda h(\tau; u).$$

## The rough Heston Riccati equation

Substitution into the convolution Riccati equation:

$$g(\tau; u) = \frac{1}{2}(u^2 - u) + \rho u h(\tau; u) + \frac{1}{2} h(\tau; u)^2, \quad \tau \geq 0$$

gives the rough Heston fractional ODE

$$D^\alpha h(\tau; u) = \frac{1}{2} u(u-1) - (\lambda - \rho \nu u) h(\tau; u) + \frac{1}{2} \nu^2 h(\tau; u)^2,$$

originally derived by El Euch and Rosenbaum.

## Two limits again

- $\lambda = 0$ :  $\kappa(\tau) = \frac{\nu}{\Gamma(\alpha)} \tau^{\alpha-1}$

$$D^\alpha h(\tau; u) = \frac{1}{2} u(u-1) + \rho \nu u h(\tau; u) + \frac{1}{2} \nu^2 h(\tau; u)^2,$$

as before.

- $\alpha = 1: \kappa(\tau) = \nu e^{-\lambda \tau}$   
(2)

$$\frac{d}{dt} h(\tau; u) = \frac{1}{2} u (u - 1) - (\lambda - \rho \nu u) h(\tau; u) + \frac{1}{2} \nu^2 h(\tau; u)^2,$$

consistent with the classical Heston derivation in (for example) Chapter 2 of [The Volatility Surface]<sup>[8]</sup>.

## Solution of the classical Heston Riccati equation

We rewrite equation (2) in the form

(3)

$$\partial_\tau h = \frac{1}{2} \eta^2 (h - r_+)(h - r_-)$$

where

$$\begin{aligned} r_\pm &= \frac{\lambda - \rho \eta u \pm \sqrt{\lambda^2 + u(\eta^2 - 2\rho \eta \lambda) - u^2 \eta^2 (1 - \rho^2)}}{\eta^2} \\ &=: \frac{\lambda - \rho \eta u \pm d}{\eta^2}. \end{aligned}$$

Integrating (3) with the terminal condition  $h(0; u) = (\kappa \star g)(0, u) = 0$  gives

$$h(\tau; u) = r_- \frac{1 - e^{-d \tau}}{1 - \frac{r_-}{r_+} e^{-d \tau}}.$$

## The Heston characteristic function

The characteristic function is given by

$$\varphi_t(T; a) = \mathbb{E}_t [e^{iaX_T}] = \exp \left\{ iaX_t + \int_t^T \xi_t(s) g(T - s; ia) ds \right\}$$

Recall that in the Heston model,

$$\xi_t(s) = (V_t - \bar{V}) e^{-\lambda(s-t)} + \bar{V}.$$

First note that

$$\int_t^T e^{-\lambda(s-t)} g(T - s; ia) ds = h(T - t; ia).$$

Also,

$$g(\tau; u) = \partial_\tau h(\tau; u) + \lambda h(\tau; u)$$

so

$$\int_t^T g(T-s; ia) ds = h(T-t; ia) + \lambda \int_t^T h(T-s; ia) ds.$$

Identifying  $h(\tau; ia)$  with  $D(a, \tau)$  and with

$$C(a, \tau) = \lambda \int_t^T h(T-s; ia) ds,$$

we obtain

(4)

$$D(a, \tau) = r_- \frac{1 - e^{-d\tau}}{1 - \frac{r_-}{r_+} e^{-d\tau}}$$

$$C(a, \tau) = \lambda \left\{ r_- \tau - \frac{2}{\eta^2} \log \left( \frac{1 - \frac{r_-}{r_+} e^{-d\tau}}{1 - \frac{r_-}{r_+}} \right) \right\}$$

as in equation (2.12) of [The Volatility Surface]<sup>[7]</sup>.

## The Heston characteristic function

For emphasis, the Heston characteristic function is given by

### The Heston characteristic function

$$\begin{aligned} \varphi_t(T; a) &= \mathbb{E}_t [e^{iaX_T}] \\ &= \exp\{iaX_t + C(a, T-t) \bar{V} + D(a, T-t) V_t\} \end{aligned}$$

## The Heston density

The Heston probability density function may be computed by Fourier inversion as

(5)

$$\begin{aligned}
 p(y, \tau) &= \int_{-\infty}^{\infty} \frac{du}{2\pi} e^{-i u y} \varphi(u) \\
 &= \frac{1}{\pi} \int_0^{\infty} du \operatorname{Re} \left[ \exp \{ C(u, \tau) \bar{V} + D(u, \tau) V - i u y \} \right]
 \end{aligned}$$

- Note that  $y = \log S_T / S_t$  is the forward variable.

## Information content of the characteristic function

The following are equivalent in the sense that given one of these, all of the others may be derived:

- All  $T$  expiration European option prices.
- The  $T$  expiration risk-neutral density.
- The  $T$  expiration implied volatilities.
- The  $T$  expiration characteristic function.

## R-implementation of the Heston characteristic function

In [1]: `# Download .zip from GitHub`

```

release_url <- "https://github.com/jgatheral/MTH9875/raw/main/9875-3.zip"
local_zip <- "9875-3.zip"

download.file(release_url, local_zip, mode = "wb")
unzip(local_zip)

```

In [2]: `source("black_scholes.R")`  
`source("lewis.R")`  
`source("heston.R")`  
`source("rough_heston_pade.R")`

In [3]: `phiHeston`

```

function (params)
{
  lambda <- params$lambda
  rho <- params$rho
  eta <- params$eta
  vbar <- params$vbar
  v <- params$v
  function(u, t) {
    al <- -u * u/2 - (0+1i) * u/2
    bet <- lambda - rho * eta * (0+1i) * u
    gam <- eta^2/2
    d <- sqrt(bet * bet - 4 * al * gam)
    rp <- (bet + d)/(2 * gam)
    rm <- (bet - d)/(2 * gam)
    g <- rm/rp
    D <- rm * (1 - exp(-d * t))/(1 - g * exp(-d * t))
    C <- lambda * (rm * t - 2/eta^2 * log((1 - g * exp(-(d *
      t)))/(1 - g)))
    return(exp(C * vbar + D * v))
  }
}

```

## A digression: the complex logarithm in (4)

In Heston's original paper and in most other papers on the subject,  $C(u, \tau)$  is written (almost) equivalently as

(6)

$$C(u, \tau) = \lambda \left\{ r_+ \tau - \frac{2}{\eta^2} \log \left( \frac{e^{+d\tau} - g}{1 - g} \right) \right\}$$

with  $g = \frac{r_-}{r_+}$ .

The reason for the qualification "almost" is that this definition coincides with our previous one only if the imaginary part of the complex logarithm is chosen so that  $C(u, \tau)$  is continuous with respect to  $u$ .

It turns out that taking the principal value of the logarithm in (6) causes  $C(u, \tau)$  to jump discontinuously each time the imaginary part of the argument of the logarithm crosses the negative real axis.

## BCC parameters

[Bakshi, Cao and Chen]<sup>[3]</sup> found (more or less) the following Heston (SV) parameters in a fit of the Heston model to historical volatility surfaces:

Table 1: BCC parameters

|           |       |
|-----------|-------|
| $v$       | 0.04  |
| $\bar{v}$ | 0.04  |
| $\lambda$ | 1.15  |
| $\eta$    | 0.39  |
| $\rho$    | -0.64 |

## A numerical example with BCC parameters

Recall the two versions of  $C(u, \tau)$ :

$$C^{JG}(u, \tau) = \lambda \left\{ r_- \tau - \frac{2}{\eta^2} \log \left( \frac{1 - g e^{-d \tau}}{1 - g} \right) \right\}$$

and

$$C^{Heston}(u, \tau) = \lambda \left\{ r_+ \tau - \frac{2}{\eta^2} \log \left( \frac{e^{+d \tau} - g}{1 - g} \right) \right\}.$$

We code these below.

```
In [4]: bigC <- function(params){
  lambda <- params$lambda
  rho <- params$rho
  eta <- params$eta
  vbar <- params$vbar
  v <- params$v

  fj <- function(u, t, j){
    al <- -u*u/2 - 1i*u/2 + 1i*j*u
    bet <- lambda - rho*eta*1i*u-rho*eta*j
    gam <- eta^2/2
    d <- sqrt(bet*bet - 4*al*gam)
    rp <- (bet + d)/(2*gam)
    rm <- (bet - d)/(2*gam)
    g <- rm / rp
    C.JG <- lambda * (rm * t - 2/eta^2 * log( (1 - g*exp(-(d*t)))/(1 - g)))
    C.Heston <- lambda * (rp * t - 2/eta^2 * log( (exp(d*t) - g)/(1 - g)))
    return(list(C.JG=C.JG,C.Heston=C.Heston))
  }
}
```



```
}
}
```

Check the two versions with  $\tau = 3$ , and with BCC parameters:

```
In [5]: paramsBCC <- list(lambda = 1.15, rho = -0.64, eta = 0.39, vbar = 0.04, v = .04)
```

```
In [6]: bigC(paramsBCC)(u=10, t=3, j=0)
```

**\$C.JG** -47.5424985325402+26.838817444334i

**\$C.Heston** -47.54249853254+121.850824061777i

The imaginary parts are very different. Which function works better?

## Comparison plot

```
In [7]: library(repr)
options(repr.plot.width=10, repr.plot.height=7, repr.plot.res=200)
```

```
In [8]: u.vec <- seq(0, 10, .01)
x <- bigC(paramsBCC)(u=u.vec, t=3, j=0)
```

```
In [9]: plot(Re(x$C.Heston), Im(x$C.Heston), type="l", col="red", lty=2, lwd=2, xlab="Re(C)",
lines(Re(x$C.JG), Im(x$C.JG), type="l", col="green4", lwd=1.5))
```

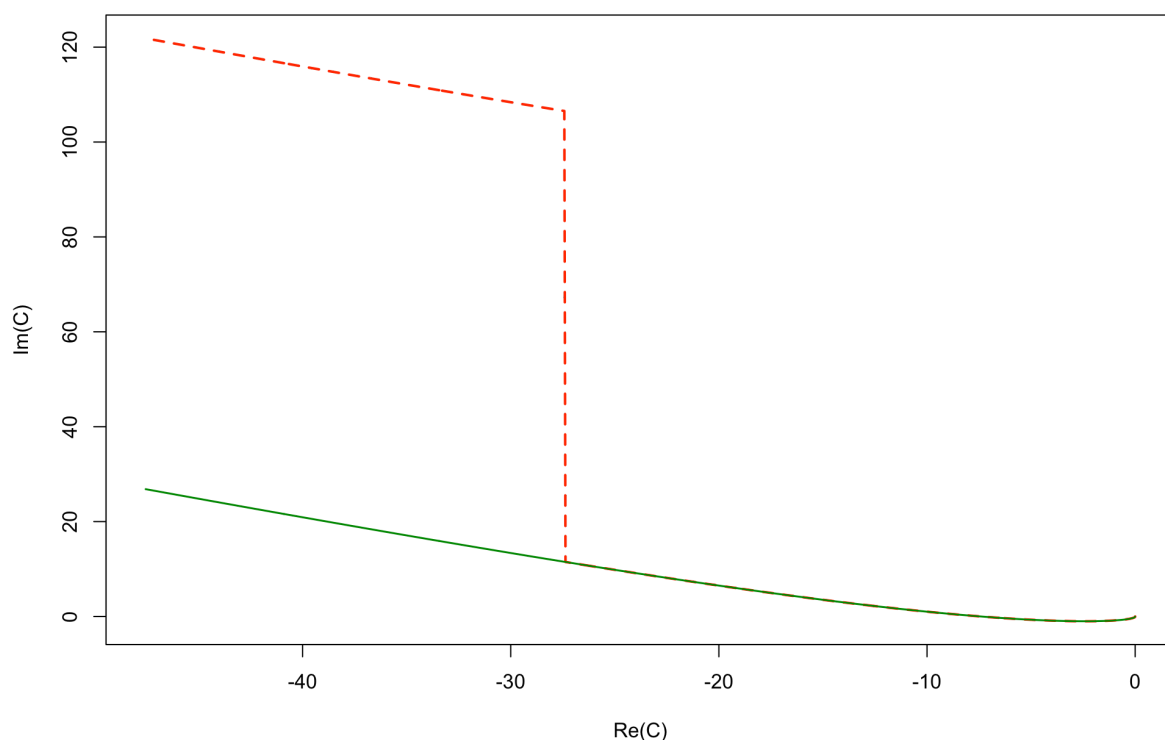


Figure 1:  $C^{Heston}(\cdot)$  in red,  $C^{JG}(\cdot)$  in green.

## Solutions in the literature

One conventional resolution is to keep careful track of the winding number in the integration (14) so as to remain on the same Riemann sheet.

This leads to practical implementation problems because standard numerical integration routines cannot be used. The paper of [Kahl and Jäckel]<sup>[10]</sup> concerns itself with this problem and provides an ingenious resolution.

## A better solution

- With *our* definition (4) of  $C(u, \tau)$ , whenever the imaginary part of the argument of the logarithm is zero, the real part is positive.
  - Plotted in the complex plane, the argument of the logarithm never cuts the negative real axis.
  - This result was proved by [Albrecher, Mayer, Schoutens and Tistaert]<sup>[2]</sup>.
- It follows that with our definition of  $C(u, \tau)$ , taking the principal value of the logarithm leads to a continuous integrand over the full range of integration.

## The rough Heston characteristic function

- There exist a number of standard numerical techniques, such as the Adams scheme, for solving fractional differential equations such as the rough Heston fractional Riccati equation.
  - These techniques are all slow!
- [Gatheral and Radoičić]<sup>[11][12]</sup> showed how to approximate the solution of the rough Heston fractional Riccati equation by a rational function.
  - This approximation solution is just as fast as the classical Heston solution and appears to be more accurate than the Adams scheme for any reasonable number of time steps!

## Solving the Riccati equation in the fractional case

For any affine forward variance model, the CGF is  
(1)

$$\varphi_t^T(a) := \log \mathbb{E} [e^{i a X_{t,T}} | \mathcal{F}_t] = \int_t^T \xi_t(s) g(T - s; a) ds$$

where  $g$  solves the convolution Riccati equation

$$g = -\frac{1}{2} a (a + i) + \rho a i (\kappa \star g) + \frac{1}{2} (\kappa \star g)^2.$$

- For the classical Heston model ( $\alpha = 1$ ) we solved this in closed form.
- For  $\alpha < 1$  there is no closed form.

## The convolution Riccati equation as a fractional ODE

With the Mittag-Leffler kernel  $\kappa(\tau) = \nu \tau^{\alpha-1} E_{\alpha,\alpha}(-\lambda \tau^\alpha)$  and  $h := (\kappa \star g)/\nu$ , the convolution Riccati equation becomes:

### The rough Heston Riccati equation (El Euch and Rosenbaum)

$$D^\alpha h(t; a) = -\frac{1}{2} a (a + i) + (i \rho \nu a - \lambda) h(t; a) + \frac{1}{2} \nu^2 h^2(t; a); \quad (3)$$

$$I^{1-\alpha} h(t; a) = 0. \quad (4)$$

- The Adams scheme solves this numerically.
  - However, too slow for calibration.
- We build global Padé (rational) approximations.
  - We match the true solution for short times and long times.

## Short times

Try  $h(t; a) = \sum_{j \geq 1} b_j t^{j\alpha}$ . Substituting and matching powers of  $t$ ,

$$b_1 = -\frac{1}{\Gamma(1+\alpha)} \frac{1}{2} a (a + i)$$

$$b_2 = \frac{\Gamma(1+\alpha)}{\Gamma(1+2\alpha)} (i \rho a - \lambda') \nu b_1$$

with  $\lambda' = \lambda/\nu$ , and in general

$$b_k = \frac{\Gamma(1+(k-1)\alpha)}{\Gamma(1+k\alpha)} \left\{ -\tilde{\lambda} \nu b_{k-1} + \frac{1}{2} \nu^2 \sum_{i+j=k-1} b_i b_j \right\}, \quad \tilde{\lambda} = \lambda' - i \rho a.$$

## Long times

Rewrite the Riccati equation as

$$D^\alpha h(t; a) = \frac{1}{2} (\nu h(t; a) - r_-) (\nu h(t; a) - r_+),$$

with  $r_\pm = \lambda' - i\rho a \pm A$ ,  $A = \sqrt{a(a+i) + (\lambda' - i\rho a)^2}$ .

Now try  $h(t; a) = \sum_{k \geq 0} g_k t^{-k\alpha}$ . Matching powers of  $t$ ,

$$g_0 = \frac{r_-}{\nu}, \quad g_1 = -\frac{1}{A\nu} \frac{1}{\Gamma(1-\alpha)} g_0,$$

again with a recursion for the higher  $g_k$ .

## The rational (or Padé) approximation

- We have the behavior of  $h$  for both short time and long times.
- The only global rational approximations matching *both* expansions are the diagonal ones:  
(2)

$$h^{(n,n)}(t; a) = \frac{\sum_{i=1}^n p_{n,i} y^i}{\sum_{j=0}^n q_{n,j} y^j}, \quad y = \nu t^\alpha.$$

- For  $n = 2$  there are four unknowns, fixed by four equations:  $b_1, b_2$  from the short-time expansion and  $g_0, g_1$  from the long-time expansion.

## How good are the Padé approximations?

- $h^{(3,3)}$ ,  $h^{(4,4)}$  and  $h^{(5,5)}$  are essentially indistinguishable from a 1,000-step Adams scheme.
- All of them are just as fast to compute as the *classical* Heston solution.
- Approximation quality *improves* as  $H$  decreases
  - exactly where we estimate  $H$  either from data or the volatility surface.
- When  $H = \frac{1}{2}$ , we know the characteristic function is classical Heston.
  - We can measure the approximation error against this closed form.

## Convergence of the Padé approximations

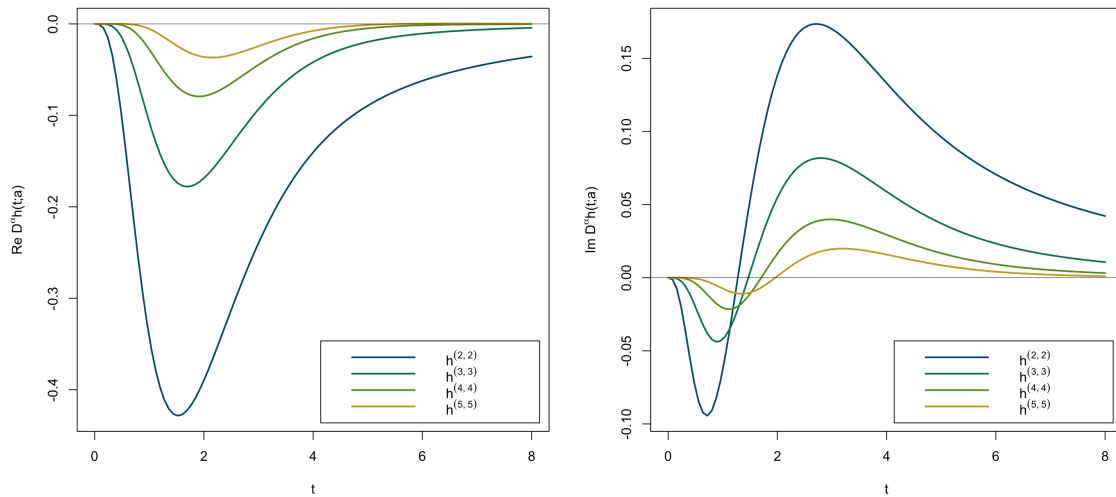


Figure 2: Approximation errors for  $\Re D^\alpha h(t; 3 - i/2)$  (left) and  $\Im D^\alpha h(t; 3 - i/2)$  (right).

## Convergence of the Padé approximations

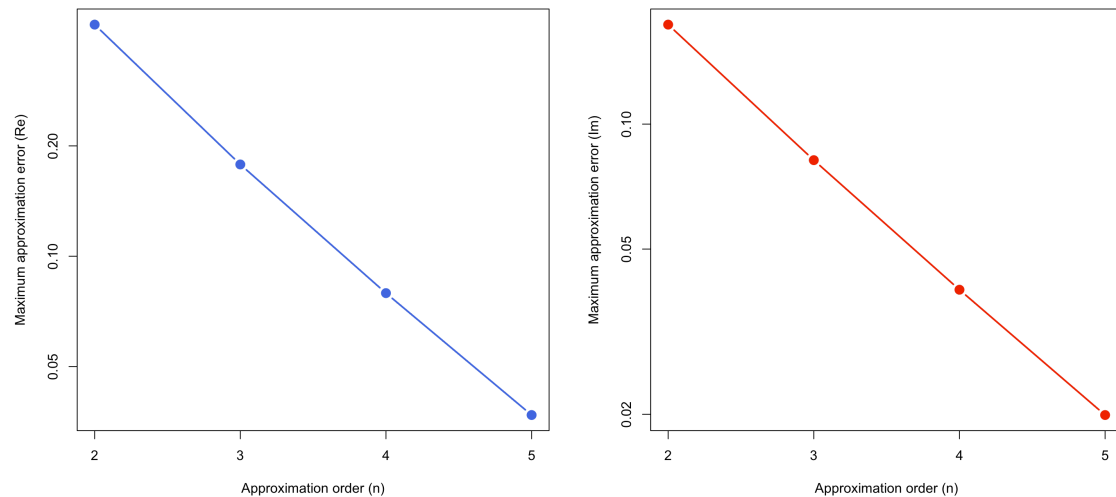


Figure 3: Maximum absolute approximation errors for  $\Re D^\alpha h(t; 3 - i/2)$  (left) and  $\Im D^\alpha h(t; 3 - i/2)$  (right) as a function of the order  $n$  of the Padé approximation  $h(n, n)$ . The y-axis scale is logarithmic.

## Which order should we use?

- Convergence in the order  $n$  appears to be exponential.
- But the coefficients of higher-order approximants are harder to derive, and the resulting functions are slower to evaluate.

- So even if  $h^{(7,7)}$  were more accurate than  $h^{(4,4)}$ , it would be slower;  $h^{(4,4)}$  is the practical choice.
- The code is in `rough_heston_pade.R` in this lecture's zip.

## Computing option prices from the characteristic function

It turns out (see [Carr and Madan]<sup>[5]</sup> and [Lewis]<sup>[13]</sup>) that it is quite straightforward to get option prices by inverting the characteristic function of a given stochastic process (if it is known in closed-form).

The formula we will use is a special case of formula (2.10) of Lewis (as usual we assume zero interest rates and dividends):

### Formula (2.10) of Lewis

(7)

$$C(S, K, t, T) = S - \sqrt{SK} \frac{1}{\pi} \int_0^\infty \frac{du}{u^2 + \frac{1}{4}} \operatorname{Re} [e^{-iuk} \varphi_t(T; u - i/2)]$$

with  $k = \log\left(\frac{K}{S}\right)$ . We now proceed to prove this formula.

## Heston greeks

It is worth noting that taking derivatives of the Heston option value with respect to  $x$  or  $v$  in order to compute delta, gamma, vega and so on is extremely straightforward because the functions  $C(u, \tau)$  and  $D(u, \tau)$  are independent of  $x$  and  $v$ .

## Proof of (7)

A covered call position has the payoff  $\min[S_T, K]$  where  $S_T$  is the stock price at time  $T$  and  $K$  is the strike price of the call.

Consider the Fourier transform of this covered call position  $G(k, \tau)$  with respect to the log-strike  $k := \log(K/F)$  defined by

$$\hat{G}(u, \tau) = \int_{-\infty}^{\infty} e^{i u k} G(k, \tau) dk$$

Denoting time-to-expiration by  $\tau = T - t$ , and setting interest rates and dividends to zero as usual, we have that

$$\begin{aligned} \frac{1}{S} \hat{G}(u, \tau) &= \int_{-\infty}^{\infty} e^{i u k} \mathbb{E} [\min[e^{x_\tau}, e^k]^+] dk \\ &= \mathbb{E} \left[ \int_{-\infty}^{\infty} e^{i u k} \min[e^{x_\tau}, e^k]^+ dk \right] \\ &= \mathbb{E} \left[ \int_{-\infty}^{x_\tau} e^{i u k} e^k dk + \int_{x_\tau}^{\infty} e^{i u k} e^{x_\tau} dk \right] \\ &= \mathbb{E} \left[ \frac{e^{(1+iu)x_\tau}}{1+iu} - \frac{e^{(1+iu)x_\tau}}{iu} \right] \text{ only if } 0 < \text{Im}[u] < 1! \\ &= \frac{1}{u(u-i)} \mathbb{E} [e^{(1+iu)x_\tau}] \\ &= \frac{1}{u(u-i)} \varphi(\tau; u-i) \end{aligned}$$

by definition of the characteristic function  $\varphi(\tau; u)$ .

- Note that the transform of the covered call value exists only if  $0 < \text{Im}[u] < 1$ .
  - It is easy to see that this derivation would go through pretty much as above with other payoffs.
  - The region where the transform exists depends on the payoff.

To get the call price in terms of the characteristic function, we express it in terms of the covered call and invert the Fourier transform, integrating along the line  $\text{Im}[u] = 1/2$ .

Then

$$\begin{aligned}
C(S, K, T) &= S - S \frac{1}{2\pi} \int_{-\infty+i/2}^{\infty+i/2} \frac{du}{u(u-i)} \varphi(T; u-i) e^{-iku} \\
&= S - S \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{du}{(u+i/2)(u-i/2)} \varphi(T; u-i/2) e^{-ik(u+i/2)} \\
&= S - \sqrt{SK} \frac{1}{\pi} \int_0^{\infty} \frac{du}{u^2 + \frac{1}{4}} \operatorname{Re} [e^{-iuk} \varphi(T; u-i/2)]
\end{aligned}$$

with  $k = \log\left(\frac{K}{S}\right)$ .

## R implementation of the Lewis formula

In [10]: option.OTM.raw

```
function (phi, k, tau)
{
  integrand <- function(u) {
    Re(exp(-(0 + (0+1i)) * u * k) * phi(u - (0 + (0+1i))/2,
      tau)/(u^2 + 1/4))
  }
  k.minus <- (k < 0) * k
  res <- exp(k.minus) - exp(k/2)/pi * integrate(integrand,
    lower = 0, upper = Inf, rel.tol = 1e-10, subdivisions = 1000)
  $value
  return(ifelse(res < 0, NA, res))
}
```

- Note that there is only one numerical integration here.
- In the version presented in Chapter 2 of [The Volatility Surface]<sup>[7]</sup>, two integrations are required.
  - The option price is then given by the difference between two large numbers, introducing numerical error.

## The Heston model in R

- In the next few slides, we will exhibit an implementation of the Heston model in R.
- We can then develop some intuition for the behavior of the model through experiment.

## Black-Scholes



First, we need an implementation of the Black-Scholes formula (this is in `black_scholes.R`).

In [11]: `BSFormula`

```
function (S0, K, T, r, sigma)
{
  x <- log(S0/K) + r * T
  sig <- sigma * sqrt(T)
  d1 <- x/sig + sig/2
  d2 <- d1 - sig
  pv <- exp(-r * T)
  return(S0 * pnorm(d1) - pv * K * pnorm(d2))
}
```

## Implied volatility computation

Then, we need an implied volatility computation (this is also in `black_scholes.R`).

In [12]: `BSImpliedVolCall`

```
function (S0, K, T, r, C)
{
  nK <- length(K)
  sigmaL <- rep(1e-10, nK)
  CL <- BSFormula(S0, K, T, r, sigmaL)
  sigmaH <- rep(10, nK)
  CH <- BSFormula(S0, K, T, r, sigmaH)
  while (mean(sigmaH - sigmaL) > 1e-10) {
    sigma <- (sigmaL + sigmaH)/2
    CM <- BSFormula(S0, K, T, r, sigma)
    CL <- CL + (CM < C) * (CM - CL)
    sigmaL <- sigmaL + (CM < C) * (sigma - sigmaL)
    CH <- CH + (CM >= C) * (CM - CH)
    sigmaH <- sigmaH + (CM >= C) * (sigma - sigmaH)
  }
  return(sigma)
}
```

## A numerical example

First generate some option prices.

```
In [13]: vols <- c(0.23,0.20,0.18)
         K <- c(0.9,1.0,1.1)
         (optVals <- BSFormula(S0=1,K,T=1,r=0,sigma=vols))
```

0.145896960390544 · 0.079655674554058 · 0.0355767789605738

Then compute implied volatilities.

```
In [14]: (impVols <- BSImpliedVolCall(S0=1, K, T=1, r=0, C=optVals))
```

0.230000000035418 · 0.200000000065986 · 0.179999999989352

## Pick some Heston parameters

```
In [15]: subHeston <- list(lambda = 0.6067, rho = -0.7571, eta = 0.2928, vbar = 0.0707, v
         option.OTM(phiHeston(subHeston), 0, 1)
         impvol.phi(phiHeston(subHeston))(0,1)
```

0.0970140580426363

0.243780488551484

## Now draw a Heston 1-year smile

```
In [18]: vol <- Vectorize(function(k){as.numeric(tryCatch(impvol.phi(phiHeston(subHes
```

```
In [19]: system.time(curve(vol(x), from=-2.3, to=1.5, col="red2"))
```

```
   user  system elapsed
0.164   0.015   0.179
```

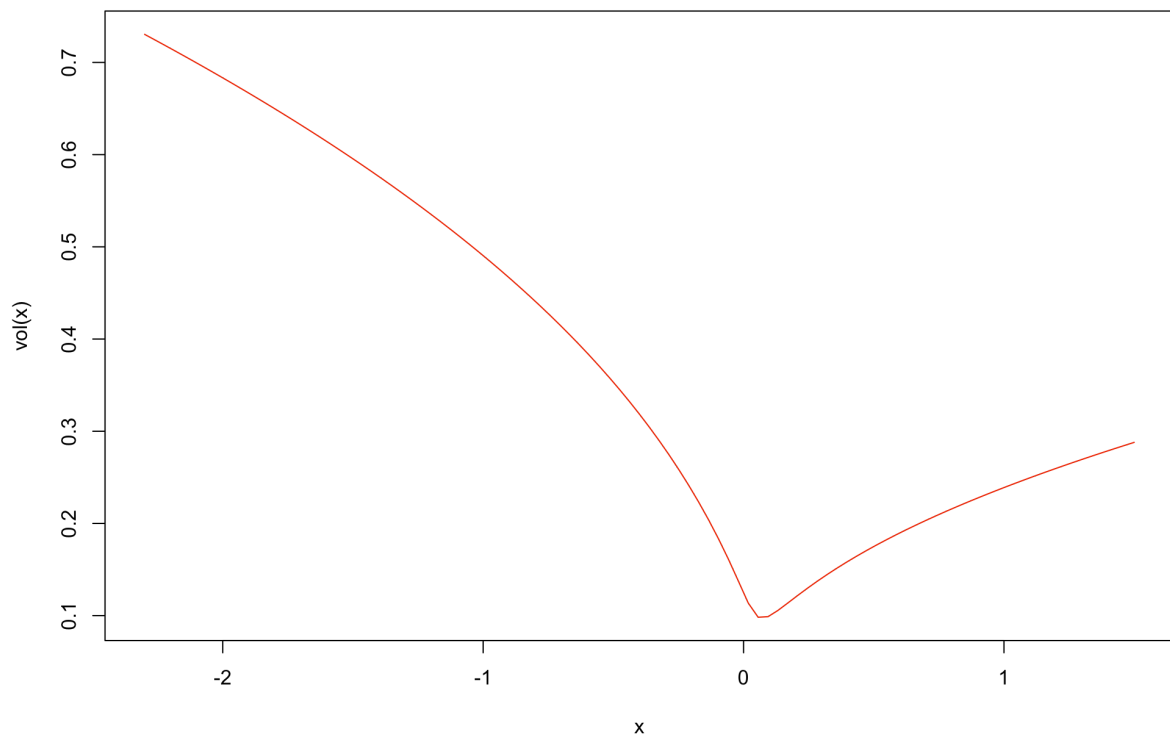


Figure 4: A one year classical Heston smile.

## Some notable features of R

- Complex arithmetic with `1i`.
- Functional programming:
  - This is what allows us to code a function which is called as:

```
impvol.phi(phiHeston(paramsBCC))(0,1)
```

- We can define a function that returns a function (and so on indefinitely).
- We could even conveniently define a new function:

```
impvolBCC <-  
  impvol.phi(phiHeston(paramsBCC))
```

- We can conceptually separate parameters and variables rather than having to carry all the parameters around with each function call.

## The Heston smile with BCC parameters

First we list the BCC parameters.

```
In [20]: paramsBCC <- list(lambda = 1.15, rho = -0.64, eta = 0.39, vbar = 0.04, v = .04)
```

Then, generate the smile as before.

```
In [21]: vol <- function(k){
  sapply(k, function(x){impvol.phi(phiHeston(paramsBCC))(x,1)}}
  system.time(curve(vol(x), from=-.4, to=.4, col="red"))
```

```
user  system elapsed
0.047  0.002  0.049
```

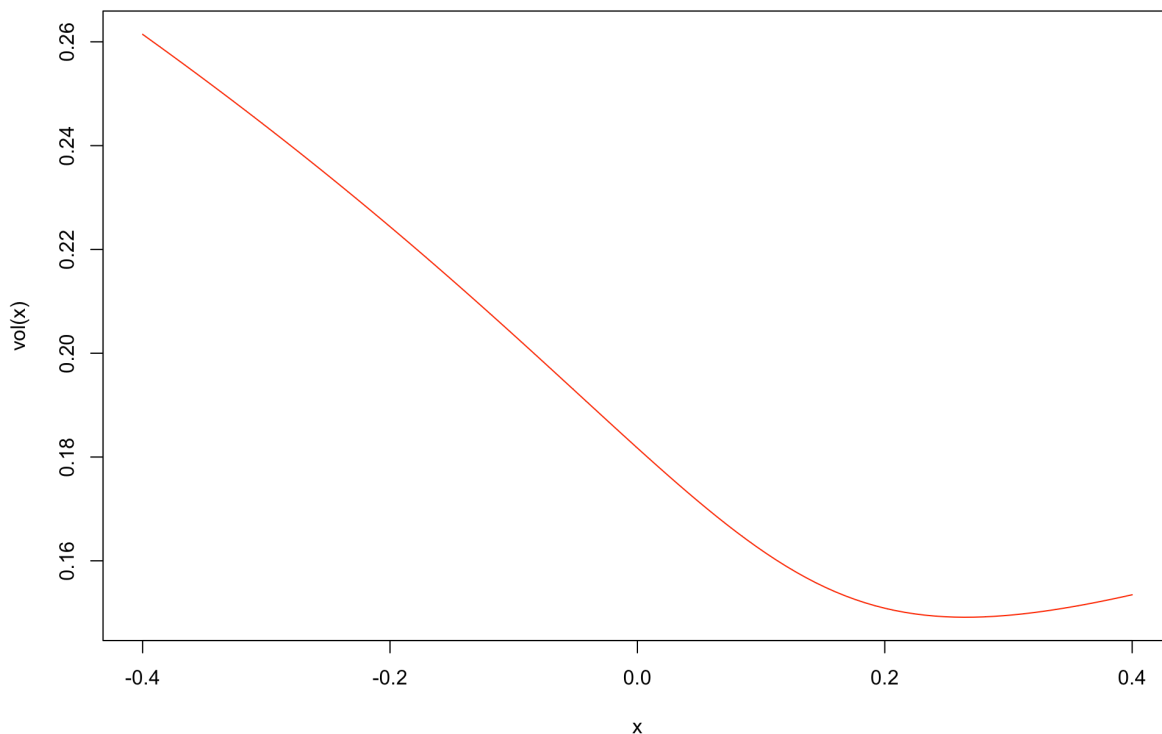


Figure 5: The  $\tau = 1$  Heston smile with BCC parameters.

## Sensitivity of the Heston smile to $\rho$

```
In [22]: subH <- function(incr){
  tmp <- paramsBCC
  tmp$rho <- tmp$rho+incr
  return(tmp)
}

vol <- function(phi)function(k){sapply(k, function(x){impvol.phi(phi)(x,1)}})
```

```
yrange <- c(0.15,.3) curve(vol(phiHeston(paramsBCC))
(x),from=-.5,to=.5,col="red",ylim=yrange,lwd=2,ylab="Implied vol.",xlab="Log-strike k") for (incr in
seq(0.1,0.5,0.1)) { curve(vol(phiHeston(subH(incr)))(x),from=-.5,to=.5,col="red",lty=2,add=T) }
```

```
In [23]: yrange <- c(0.15,.3)
```

```
kk <- seq(-.5,.5,.05)
```

```
In [24]: plot(kk,vol(phiHeston(paramsBCC))(kk),col="red2",ylim=yrange,type="l",lwd=2,
for (incr in seq(0.1,0.5,0.1))
{
  lines(kk,vol(phiHeston(subH(incr)))(kk),col="red2",lty=2)
}
```

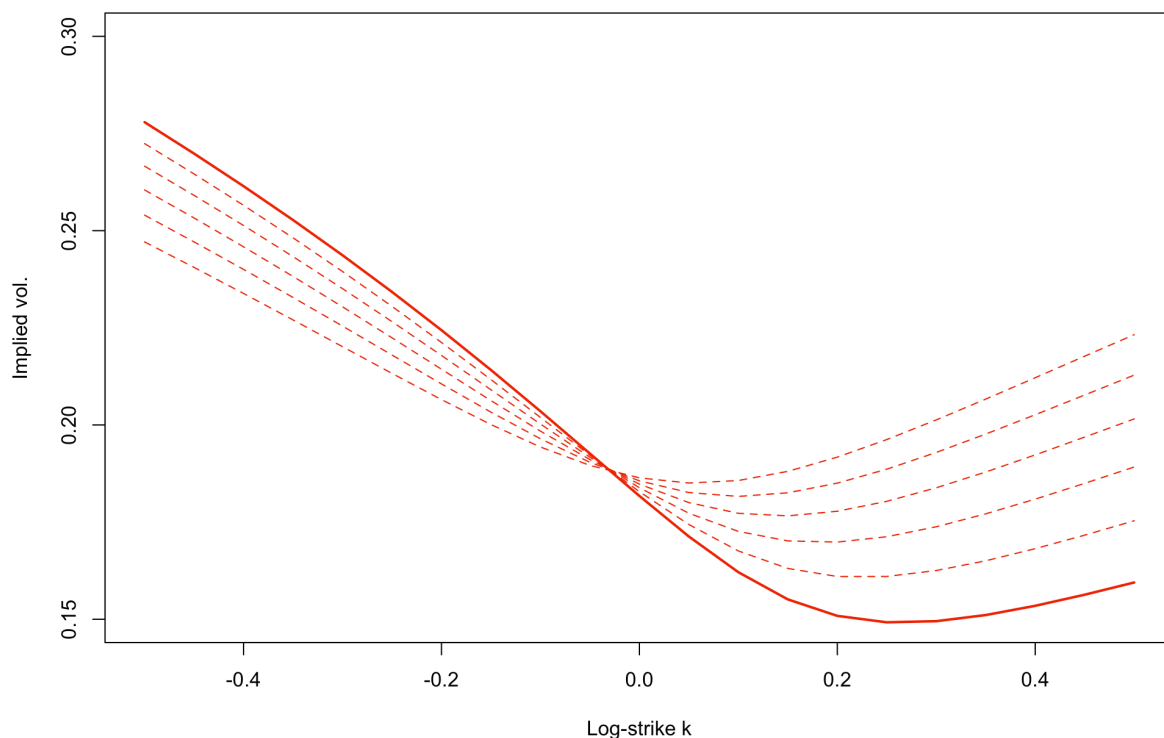


Figure 6: The dotted lines are smiles with  $\rho = \rho_{BCC} + \{0.1, 0.2, 0.3, 0.4, 0.5\}$ .

## Sensitivity of the Heston smile to $\eta$

```
In [25]: subH <- function(incr){
  tmp <- paramsBCC
  tmp$eta <- tmp$eta+incr
  return(tmp)
}
```

```
In [27]: yrange <- c(0.1,.35)
curve(vol(phiHeston(paramsBCC))(x),from=-.5,to=.5,col="red",ylim=yrange,lwd=2)
for (incr in seq(0.1,0.5,0.1))
{
  curve(vol(phiHeston(subH(incr)))(x),from=-.5,to=.5,col="red",lty=2,add=T)
}
```

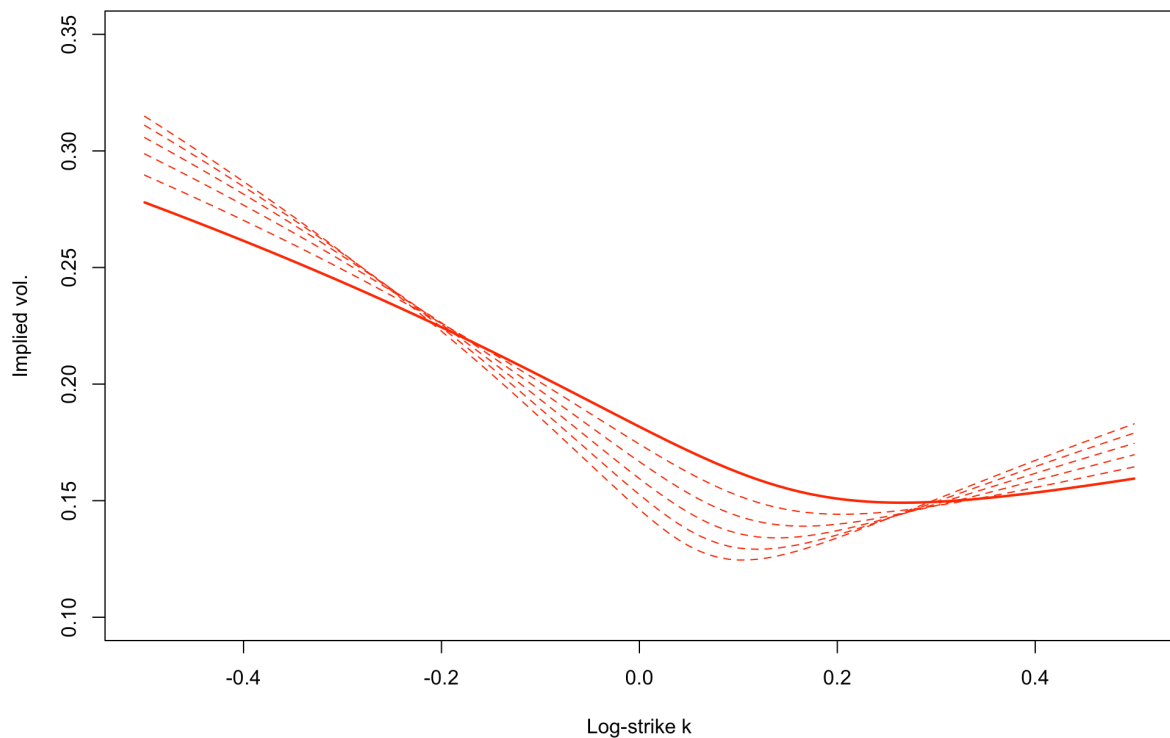


Figure 7: The dotted lines are smiles with  $\eta = \eta_{BCC} + \{0.1, 0.2, 0.3, 0.4, 0.5\}$ .

## Sensitivity of the Heston smile to $\lambda$

```
In [28]: subH <- function(incr){
  tmp <- paramsBCC
  tmp$lambda <- tmp$lambda+incr
  return(tmp)
}

In [29]: yrange <- c(0.1,.35)
plot(kk,vol(phiHeston(paramsBCC))(kk),col="red2",ylim=yrange,type="l",lwd=2,
for (incr in seq(0.5,2.5,0.5))
{
  lines(kk,vol(phiHeston(subH(incr)))(kk),col="red2",lty=2)
}
```

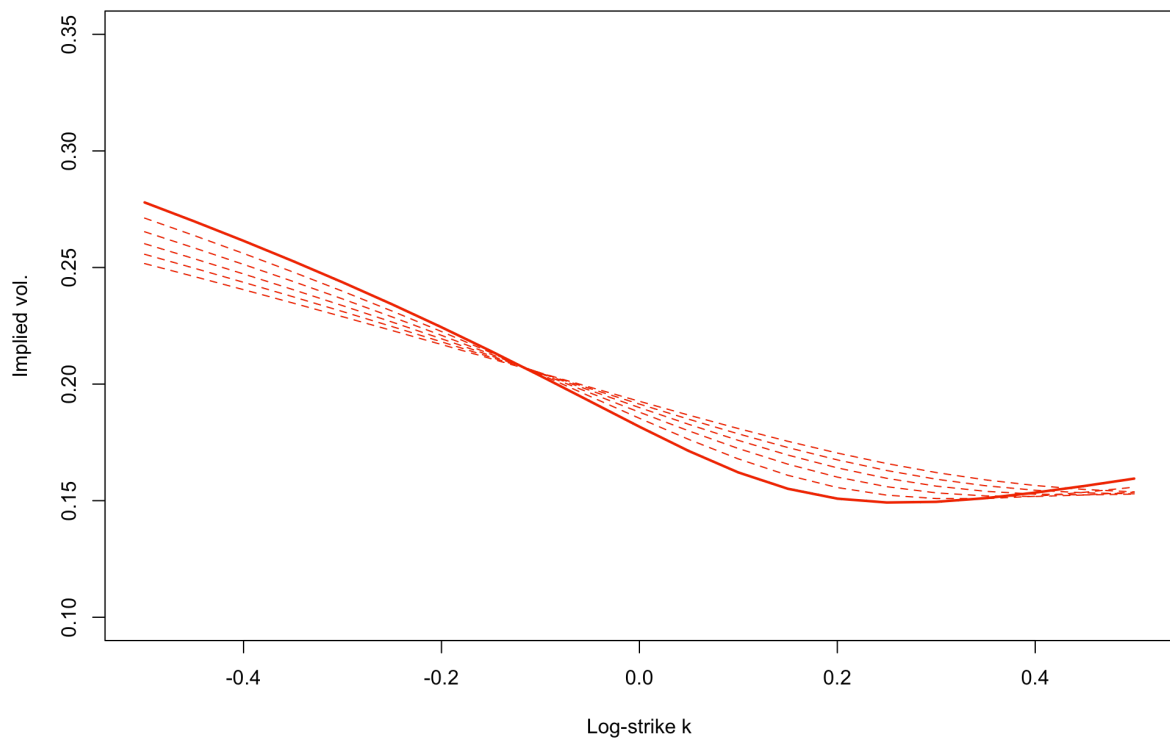


Figure 8: The dotted lines are smiles with  $\lambda = \lambda_{BCC} + \{0.5, 1.0, 1.5, 2.0, 2.5\}$ .

## Another sensitivity plot

Now increase both  $\lambda$  and  $\eta$  keeping the ratio  $\eta^2/\lambda$  constant.

```
In [30]: subH <- function(incr){
  tmp <- paramsBCC
  tmp$eta <- tmp$eta+incr
  tmp$lambda <- tmp$lambda*(tmp$eta/paramsBCC$eta)^2
  return(tmp)
}

In [31]: yrange <- c(0.1, .35)
plot(kk, vol(phiHeston(paramsBCC))(kk), col="red2", ylim=yrange, type="l", lwd=2,
  for (incr in seq(0.1, 0.5, 0.1))
  {
    lines(kk, vol(phiHeston(subH(incr)))(kk), col="red2", lty=2)
  }
```

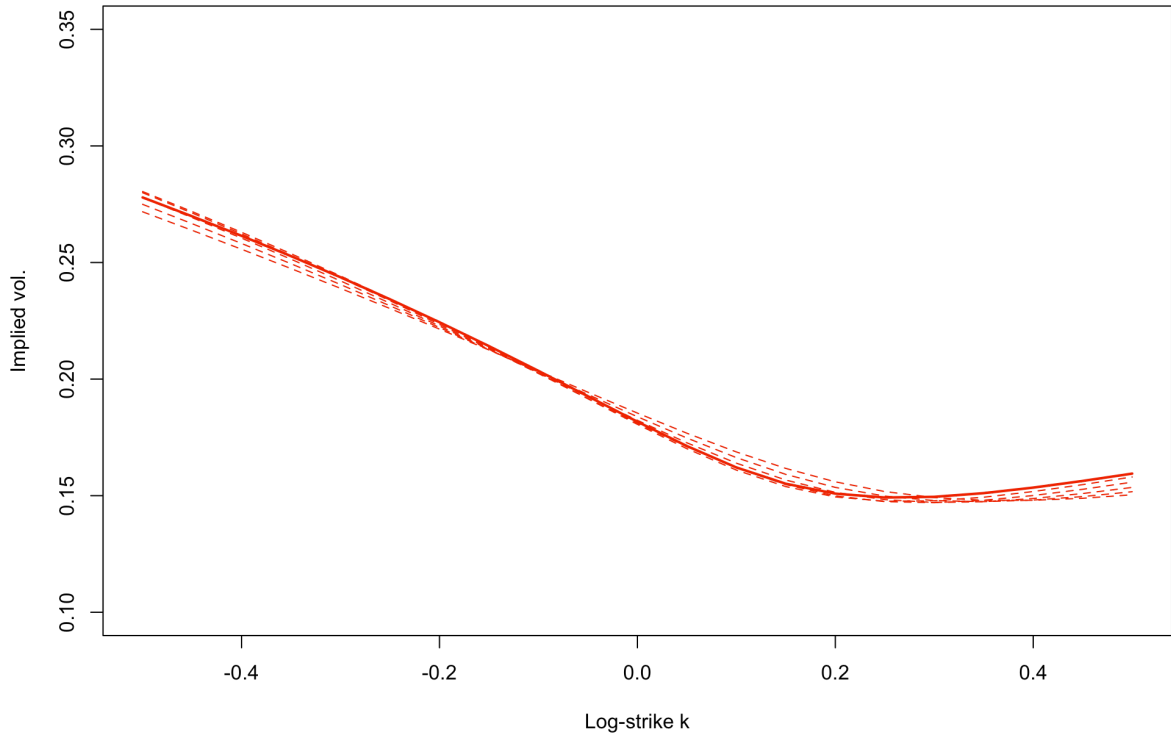


Figure 9: The dotted lines are smiles with  $\eta = \eta_{BCC} + \{0.1, 0.2, 0.3, 0.4, 0.5\}$  and  $\lambda$  incremented so as to keep the ratio  $\lambda/\eta^2$  constant.

## Variance of $V_T$ in the Heston model (version in the book)

Applying Itô's Lemma, we get

$$dV_u^2 = 2 V_u dV_u + (dV_u)^2$$

This gives

$$\mathbb{E}_t [dV_u^2] = -2 \lambda (\mathbb{E}_t [V_u^2] - \bar{V} \mathbb{E}_t [V_u]) dt + \eta^2 \mathbb{E}_t [V_u] dt$$

Also, using that  $\mathbb{E}_t [V_u] = (V_t - \bar{V}) e^{-\lambda(u-t)} + \bar{V}$ ,

$$d\mathbb{E}_t [V_u]^2 = -2 \lambda (\mathbb{E}_t [V_u]^2 - \bar{V} \mathbb{E}_t [V_u]) dt$$

So

$$d\text{var}_t [V_u] = -2 \lambda \text{var}_t [V_u] dt + \eta^2 \mathbb{E}_t [V_u] dt$$

Integrating, with  $\xi_t(u) = \mathbb{E}_t [V_u u]$ , we get

$$\text{var}_t [V_T] = \int_t^T e^{-2 \lambda (T-u)} \eta^2 \xi_t(u) du.$$



Putting  $\xi_t(u) = \bar{V}$  for simplicity, with  $\tau := T - t$ , we obtain

$$\begin{aligned}\text{var}_t[V_T] &= \eta^2 \bar{V} \int_t^T e^{-2\lambda(T-u)} du \\ &= \eta^2 \bar{V} \frac{1 - e^{-2\lambda\tau}}{2\lambda} \\ &\rightarrow \frac{\eta^2 \bar{V}}{2\lambda} \text{ as } \tau \rightarrow \infty\end{aligned}$$

- The smile depends only on the distribution of integrated variance and its correlation with the stock price, so for longer expirations, the smile should depend (roughly) only on  $\rho$  and the ratio  $\eta^2 \bar{V} / \lambda$ .

## Covariance of $V_T$ and $V_{T+\Delta}$ in affine forward variance models

Recall that  $\xi_t(s) = \mathbb{E}_t[V_s]$ .

Then

$$\begin{aligned}\text{cov}_t[V_T, V_{T+\Delta}] &= \mathbb{E}_t[V_T V_{T+\Delta}] - \mathbb{E}_t[V_T] \mathbb{E}_t[V_{T+\Delta}] \\ &= \mathbb{E}_t[\xi_T(T) \xi_T(T + \Delta)] - \xi_t(T) \xi_t(T + \Delta) \quad \text{using the tower law} \\ &= \mathbb{E}_t[(\xi_T(T) - \xi_t(T)) (\xi_T(T + \Delta) - \xi_t(T + \Delta))] \quad \text{using that } \xi_t(u) : \\ &= \mathbb{E}_t \left[ \int_t^T d\langle \xi(T), \xi(T + \Delta) \rangle_s \right] \\ &= \int_t^T \mathbb{E}_t[V_s] \kappa(T - s) \kappa(T + \Delta - s) ds \\ &= \int_t^T \xi_t(s) \kappa(T - s) \kappa(T + \Delta - s) ds\end{aligned}$$

## Variance of $V_T$ in affine forward variance models

An immediate corollary is that

$$\text{var}_t[V_T] = \int_t^T \xi_t(s) \kappa(s - t)^2 ds.$$

## Application to the Heston model

With  $\kappa(\tau) = \eta e^{-\lambda\tau}$  and putting  $\xi_t(s) = \bar{V}$  (a flat curve),  $\tau = T - t$ , we obtain

$$\begin{aligned}\mathrm{var}_t[V_T] &= \eta^2 \bar{V} \int_t^T e^{-2\lambda(s-t)} ds \\ &= \frac{\eta^2 \bar{V}}{2\lambda} \{1 - e^{-2\lambda\tau}\}.\end{aligned}$$

- The same result as before.

## Application to the rough Heston model with $\lambda = 0$

With  $\kappa(\tau) = \frac{1}{\Gamma(\alpha)} \eta \tau^{\alpha-1}$  and putting  $\xi_t(s) = \bar{V}$  (a flat curve), we obtain

$$\begin{aligned}\mathrm{var}_t[V_T] &= \frac{\eta^2 \bar{V}}{\Gamma(\alpha)^2} \int_t^T (s-t)^{2\alpha-2} ds \\ &= \frac{\eta^2 \bar{V}}{(2\alpha-1)\Gamma(\alpha)^2} \tau^{2\alpha-1} \\ &= \frac{\eta^2 \bar{V}}{2H\Gamma(\alpha)^2} \tau^{2H}\end{aligned}$$

- Variance of  $V_T$  is power-law in  $\tau = T - t$ .
- Precisely the scaling we found in Lecture 2.

## Autocorrelation of variance in the Heston model

Applying the above formula,

$$\begin{aligned}\mathrm{cov}_t[V_T, V_{T+\Delta}] &= \int_t^T \xi_t(s) \kappa(T-s) \kappa(T+\Delta-s) ds \\ &= \bar{V} \int_t^T e^{-\lambda(T-s)} e^{-\lambda(T+\Delta-s)} ds \\ &= e^{-\lambda\Delta} \bar{V} \int_t^T e^{-2\lambda(T-s)} ds.\end{aligned}$$

Thus

$$\rho(\Delta) \sim \frac{\mathrm{cov}_t[V_T, V_{T+\Delta}]}{\mathrm{var}_t[V_T]} = e^{-\lambda\Delta}.$$

## Autocorrelation of variance in the rough Heston model with $\lambda > 0$

From the affine covariance formula, with a flat forward variance curve  $\xi_t(s) = \bar{V}$  and  $u = T - s$ ,

$$\text{cov}_t[V_T, V_{T+\Delta}] = \bar{V} \int_0^\tau \kappa(u) \kappa(u + \Delta) du, \quad \tau = T - t.$$

Let  $\tau \rightarrow \infty$  to get the stationary autocorrelation:

$$\rho(\Delta) = \frac{\int_0^\infty \kappa(u) \kappa(u + \Delta) du}{\int_0^\infty \kappa(u)^2 du}.$$

## The spectral representation

- Computing  $\int_0^\infty \kappa(u) \kappa(u + \Delta) du$  directly needs Mittag-Leffler function values.
- We know the Laplace transform in closed form, so we work with it instead.
- Since  $\kappa$  vanishes on  $(-\infty, 0)$ , an autocovariance formula gives

$$\int_0^\infty \kappa(u) \kappa(u + \Delta) du = \frac{1}{\pi} \int_0^\infty |\hat{\kappa}(i\omega)|^2 \cos(\omega \Delta) d\omega.$$

- With  $\hat{\kappa}(p) = \frac{\nu}{p^{\alpha+\lambda}}$  and  $(i\omega)^\alpha = \omega^\alpha e^{i\pi\alpha/2}$ , we get

$$\rho(\Delta) \propto \int_0^\infty \frac{\cos(\omega \Delta)}{\omega^{2\alpha} + 2\lambda\omega^\alpha \cos\left(\frac{\pi\alpha}{2}\right) + \lambda^2} d\omega.$$

- When  $\alpha = 1$ ,  $\cos(\pi/2) = 0$ , and the denominator is  $\omega^2 + \lambda^2$ .
  - The integral is  $\frac{\pi}{2\lambda} e^{-\lambda\Delta}$ , proportional to the classical Heston autocorrelation function.

## Proof of the autocovariance formula

Extend  $\kappa$  by zero to  $u < 0$  and write  $R(\Delta) := \int \kappa(u) \kappa(u + \Delta) du$ . Take the Fourier transform in  $\Delta$ :

$$\begin{aligned} \int R(\Delta) e^{-i\omega\Delta} d\Delta &= \iint \kappa(u) \kappa(u + \Delta) e^{-i\omega\Delta} du d\Delta \\ &\stackrel{s = u + \Delta}{=} \underbrace{\int \kappa(u) e^{+i\omega u} du}_{\overline{\hat{\kappa}(i\omega)}} \underbrace{\int \kappa(s) e^{-i\omega s} ds}_{\hat{\kappa}(i\omega)} \\ &= |\hat{\kappa}(i\omega)|^2. \end{aligned}$$

- Equivalently,  $R = \tilde{\kappa} \star \kappa$  with  $\tilde{\kappa}(u) = \kappa(-u)$ .

Inverting the Fourier transform, we get

$$R(\Delta) = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\hat{\kappa}(i\omega)|^2 e^{i\omega\Delta} d\omega = \frac{1}{\pi} \int_0^{\infty} |\hat{\kappa}(i\omega)|^2 \cos(\omega\Delta) d\omega.$$

## R code

The following code (from Claude) computes the stationary autocorrelation of  $V$  in the rough Heston model with  $\lambda > 0$ .

```
In [32]: acf.rHeston <- function(alpha, lambda, w1=0.05, W=50, n1=600, n2=20000){
  cc <- cos(pi*alpha/2)

  # Quadrature grid: geometric near the origin (the integrand has a w^alpha
  # there), uniform beyond (we need to resolve cos(w*Delta) for Delta up to

  g <- exp(seq(log(1e-8), log(w1), length.out = n1+1))
  u <- seq(w1, W, length.out = n2+1)

  w <- c( (head(g,-1)+tail(g,-1))/2, (head(u,-1)+tail(u,-1))/2 ) # midp
  dw <- c( diff(g), diff(u) )

  wt <- dw/(w^(2*alpha) + 2*lambda*w^alpha*cc + lambda^2)/pi

  function(delta){as.numeric( cos(outer(delta, w)) %*% wt )}
}
```

## Empirical autocorrelation of variance

```
In [33]: load("rv_spx.rData")
```

```
In [34]: library(xts)
tail(rv.spx)
```

Loading required package: zoo

Attaching package: 'zoo'

The following objects are masked from 'package:base':

as.Date, as.Date.numeric

```

      [,1]
2022-06-17 2.234366e-04
2022-06-22 1.367900e-04
2022-06-23 1.499486e-04
2022-06-24 6.649679e-05
2022-06-27 9.746335e-05
2022-06-28 1.039309e-04

```

```

In [35]: v <- as.numeric(rv.spx[,1]) # Pick spx.rk
         acv <- acf(v, lag=100, plot=F)

```

## Empirical autocorrelation of variance

```

In [36]: plot(acv$lag[-1], acv$acf[-1], pch=20, ylab=expression(rho(Delta)), xlab=expression(Delta))

```

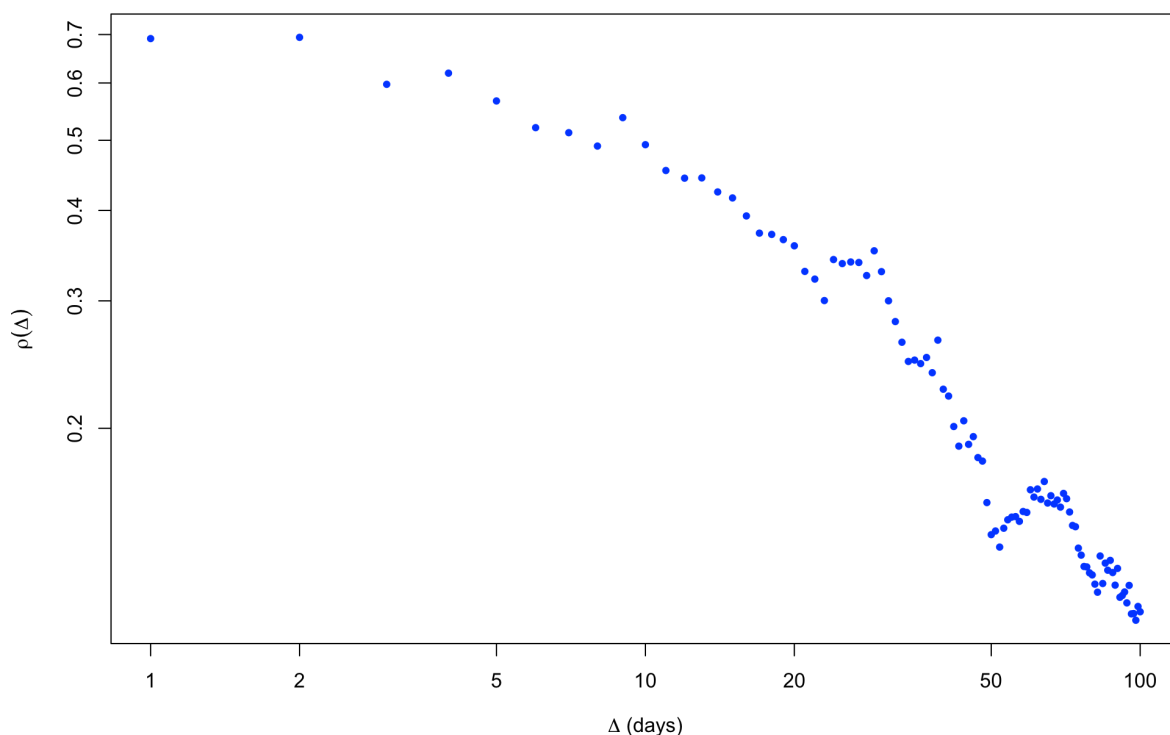


Figure 10: The empirical autocovariance function of SPX realized kernel estimates of integrated variance.

## Fitting rough Heston to the Oxford-Man data

- Three parameters: the pprefactor  $A$ , the roughness  $\alpha$ , and the mean reversion  $\lambda$ .

```

In [37]: delta <- 1:100
         rho.lv <- as.numeric(acf(v, lag.max=100, plot=FALSE)$acf)[-1]

         obj.rH <- function(p){
           model <- p[1]*acf.rHeston(alpha=p[2], lambda=p[3])(delta)

```

```

    if(any(model <= 0)) return(1e6)
    sum( (log(rho.lv) - log(model))^2 )
  }

(fit.rH <- optim(c(0.15, 0.70, 0.02), obj.rH, method="L-BFGS-B",
                lower = c(1e-4, 0.505, 1e-5),
                upper = c(10, 0.999, 2)))

alpha.hat <- fit.rH$par[2]
lambda.hat <- fit.rH$par[3]

```

```

$par      0.315942929980052 · 0.662263738203259 · 0.0563257303830174

$value     0.94669824914257

$counts    function: 102 gradient: 102

$convergence 0

$message    'CONVERGENCE: REL_REDUCTION_OF_F <= FACTR*EPSMCH'

```

## Parameter estimates

```

In [38]: data.frame(H      = alpha.hat - 1/2,
                    lambda.per.day = lambda.hat,
                    lambda.annualised = lambda.hat*252^alpha.hat,
                    timescale.days = lambda.hat^(-1/alpha.hat))

```

A data.frame: 1 × 4

| H         | lambda.per.day | lambda.annualised | timescale.days |
|-----------|----------------|-------------------|----------------|
| <dbl>     | <dbl>          | <dbl>             | <dbl>          |
| 0.1622637 | 0.05632573     | 2.193145          | 76.98361       |

- $H$  is consistent with our prior estimate in Lecture 2.
- Compare the timescale of 77 days is to 132 from squared daily log returns of SPX data in Lecture 1.

## Plot of the rough Heston ACF fit

```

In [39]: plot(delta, rho.lv, log="xy", pch=20, col="blue",
             xlab=expression(paste(Delta, " (days)")), ylab=expression(rho(Delta)),yl
             lines(delta, fit.rH$par[1]*acf.rHeston(alpha.hat, lambda.hat)(delta), col="c

```

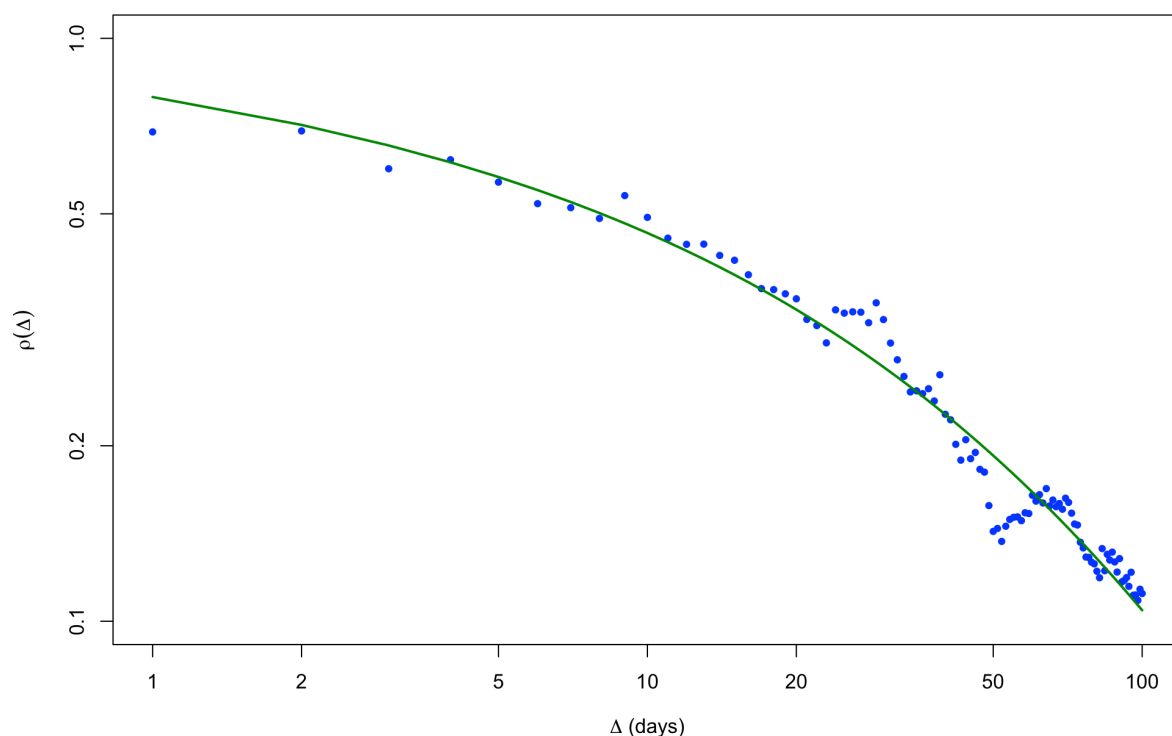


Figure 11: Empirical ACF in blue; rough Heston fit in green.

## Fitting classical Heston to the Oxford-Man data

- Two parameters: the prefactor  $A$ , and the mean reversion  $\lambda$ .

```
In [40]: delta <- 1:100
rho.lv <- as.numeric(acf(v, lag.max=100, plot=FALSE)$acf)[-1]

acf.cHeston <- function(lambda)function(delta){ exp(-lambda*delta)}

obj.cHeston <- function(p){
  model <- p[1]*acf.cHeston(lambda=p[2])(delta)
  if(any(model <= 0)) return(1e6)
  sum( (log(rho.lv) - log(model))^2 )
}

(fit.cHeston <- optim(c(0.15, 0.02), obj.cHeston, method="L-BFGS-B",
  lower = c(1e-4, .01),
  upper = c(10, .1)))

lambda.classical <- fit.cHeston$par[2]
```

```

$par      0.503539179311597 · 0.0172609129659949

$value     2.41057142205284

$counts    function: 18 gradient: 18

$convergence 0

$message    'CONVERGENCE: REL_REDUCTION_OF_F <= FACTR*EPSMCH'

```

## Parameter estimate

```

In [41]: data.frame(lambda.per.day    = lambda.classical,
  lambda.annualised = lambda.classical*252,
  timescale.days    = log(2)/lambda.classical)

```

A data.frame: 1 × 3

| lambda.per.day | lambda.annualised | timescale.days |
|----------------|-------------------|----------------|
| <dbl>          | <dbl>             | <dbl>          |
| 0.01726091     | 4.34975           | 40.15704       |

## Superimpose the rough Heston and classical Heston fits

```

In [42]: plot(delta, rho.lv, log="xy", pch=20, col="blue",
  xlab=expression(paste(Delta, " (days)")), ylab=expression(rho(Delta)),
  lines(delta, fit.rH$par[1]*acf.rHeston(alpha.hat, lambda.hat)(delta), col="red",
  lines(delta, fit.cHeston$par[1]*acf.cHeston(lambda.classical)(delta), col="green")

```

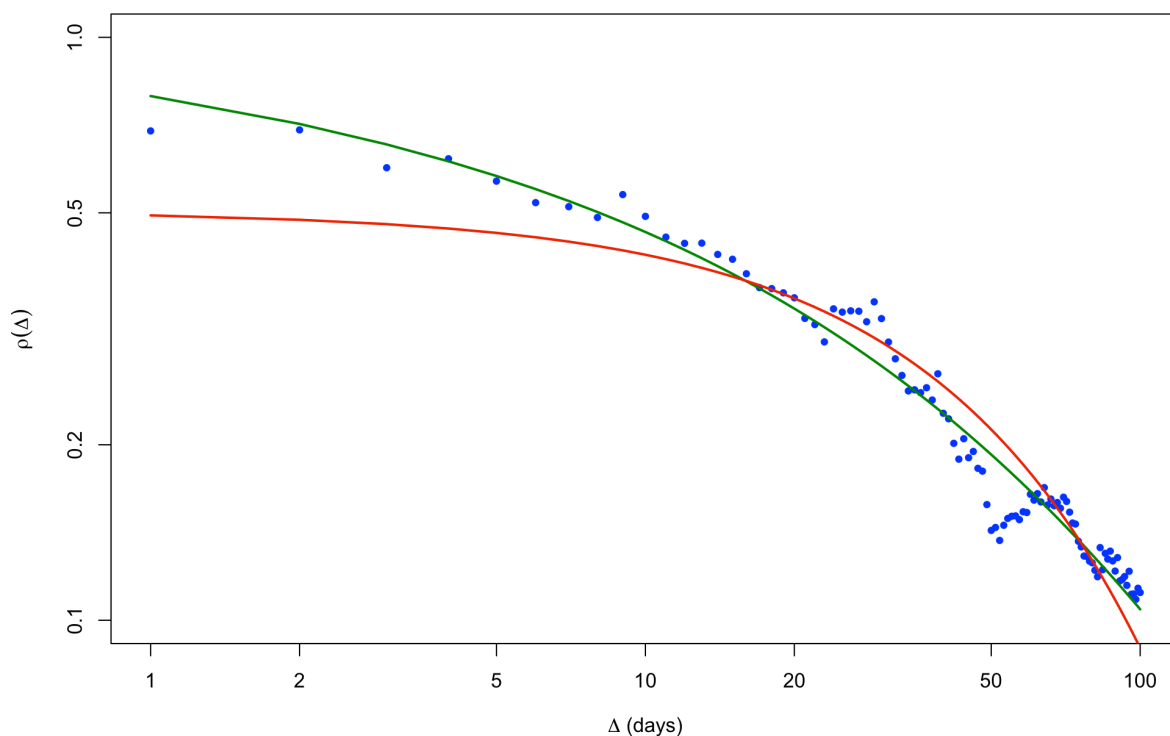




Figure 12: Empirical ACF in blue; rough Heston fit in green; classical Heston fit in red.

## Empirical autocorrelation of log-variance

```
In [43]: aclog <- acf(log(v), lag=100, plot=F)
plot(aclog$lag[-1], aclog$acf[-1], pch=20, ylab=expression(rho(Delta)), xlab=exp
```

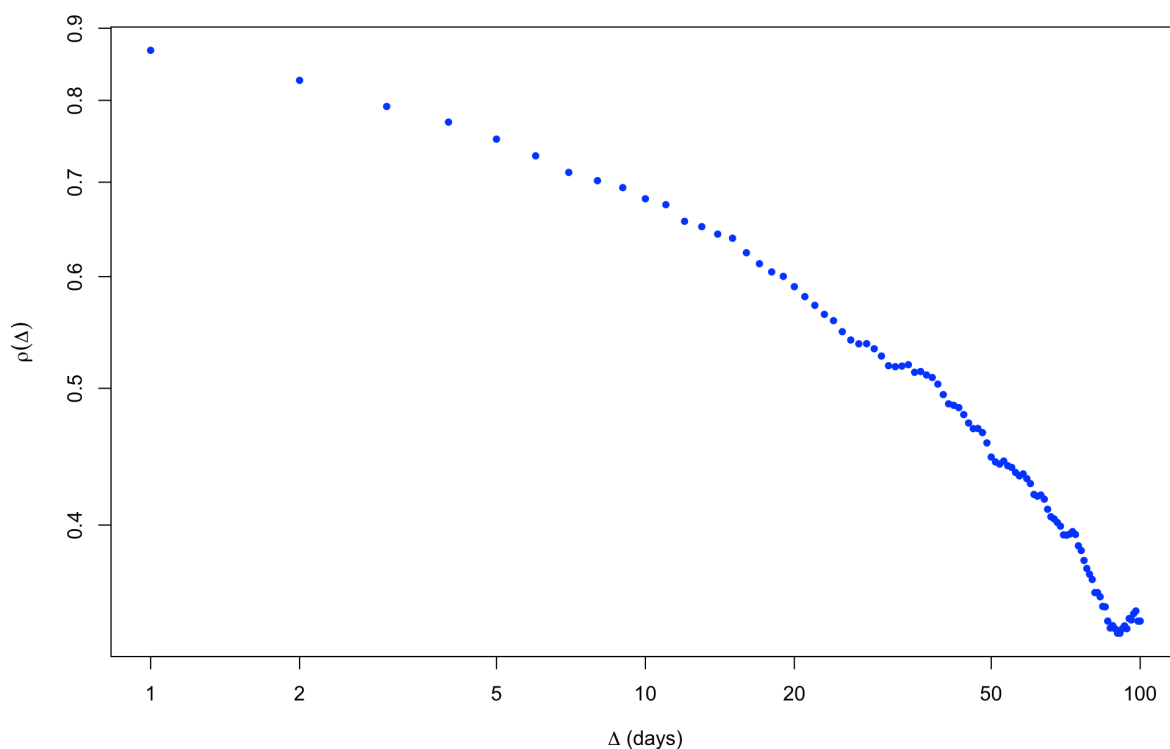


Figure 13: The empirical autocovariance function of SPX realized kernel estimates of log-integrated variance.

## Remarks

- See how much nicer Figure 13 looks compared to Figure 10!
  - This is likely because variance is close to lognormally distributed, rather than normally distributed.

## The rough Heston smile

```
In [44]: params.rHeston <- list(H=0.05, nu=0.4, rho=-.65, eta=0.4, lam=0)
xiCurve <- function(t){0.025+0*t}
```

```
In [45]: phi <- function(params){
  phiRoughHestonRational(params, xiCurve, h.approx= h.Pade44, n=100)}
```

```
In [46]: vol <- function(params)function(k){
  apply(k,function(x){impvol.phi(phi(params))(x,1)}})
kk <- seq(-.4,.4,.02)
system.time(plot(kk,vol(params.rHeston)(kk),col="red2",lwd=2,type="l"))

  user  system elapsed
1.086   0.016   1.102
```

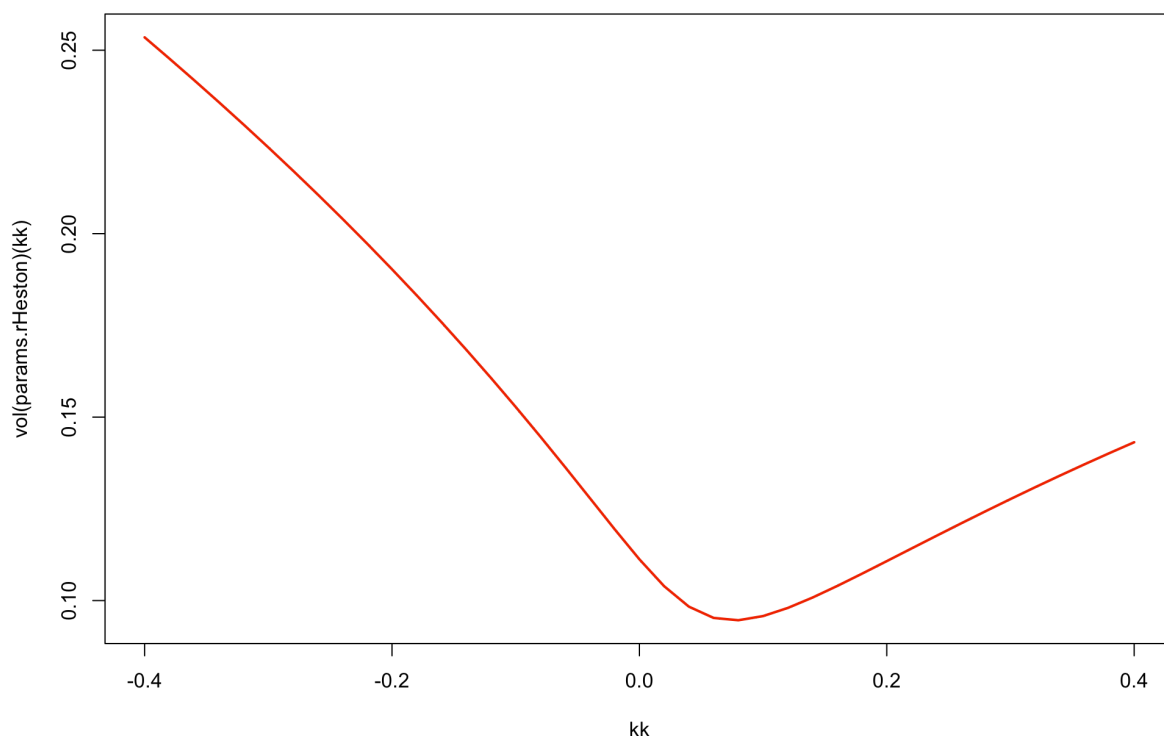


Figure 14: The rough Heston 1-year smile with  $H = 0.05$ ,  $\nu = 0.4$ ,  $\rho = -.65$ ,  $\lambda = 0$ .

## Sensitivity of the rough Heston smile to $\nu$

```
In [47]: subH <- function(incr){
  tmp <- params.rHeston
  tmp$nu <- tmp$nu+incr
  return(tmp)
}
```

```
In [48]: yrange <- c(0.05,.35)
kk <- seq(-.4,.4,.02)
plot(kk,vol(params.rHeston)(kk),col="red2",lwd=2,type="l",
     ylim=yrange,ylab="Implied vol.",xlab="Log-strike k")
for (incr in seq(0.1,0.5,0.1))
{
  lines(kk,vol(subH(incr))(kk),lty=2,col="red2")
}
```

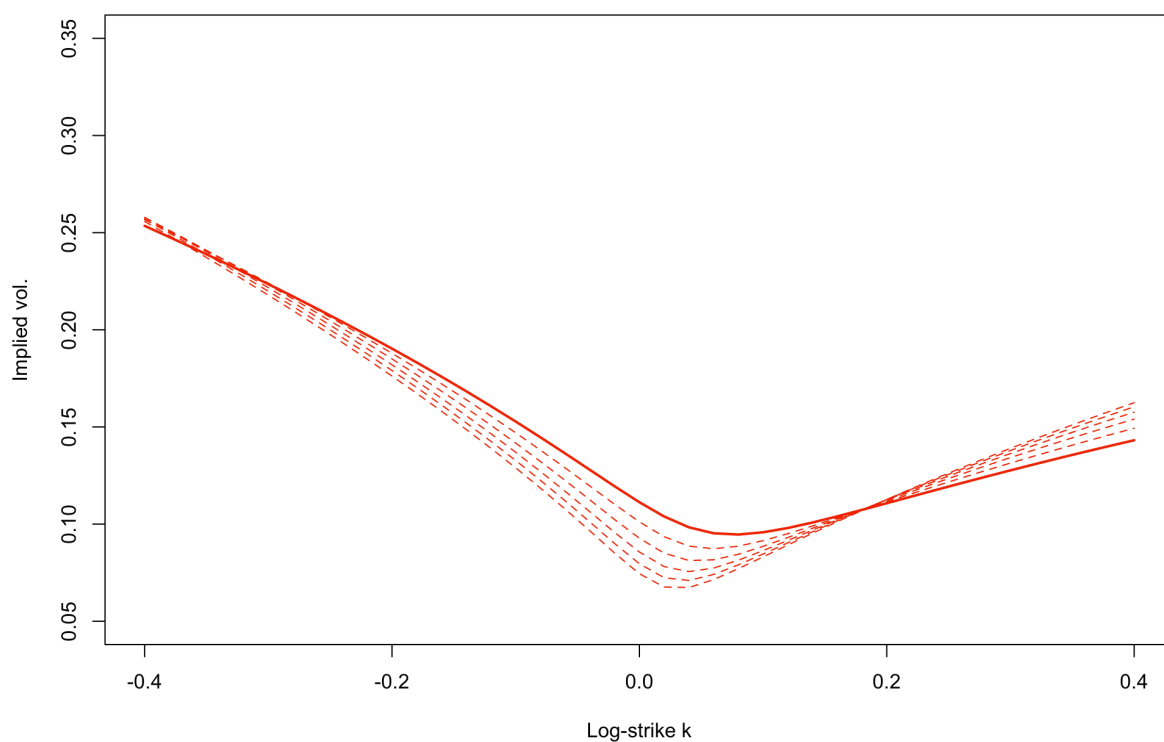


Figure 15: The dotted lines are smiles with  $\nu = \nu_0 + \{0.1, 0.2, 0.3, 0.4, 0.5\}$ .

## Sensitivity of the rough Heston 1 year smile to $H$

```
In [49]: subH <- function(incr){
  tmp <- params.rHeston
  tmp$H <- tmp$H+incr
  return(tmp)
}

In [50]: yrange <- c(0.05,.35)
kk <- seq(-.4,.4,.02)
system.time({
  plot(kk,vol(params.rHeston)(kk),col="red2",lwd=2,type="l",
    ylim=yrange,ylab="Implied vol.",xlab="Log-strike k")
  for (incr in seq(0.1,0.4,0.1))
  {
    lines(kk,vol(subH(incr))(kk),lty=2,col="red2")
  }
})

user system elapsed
5.378 0.026 5.405
```

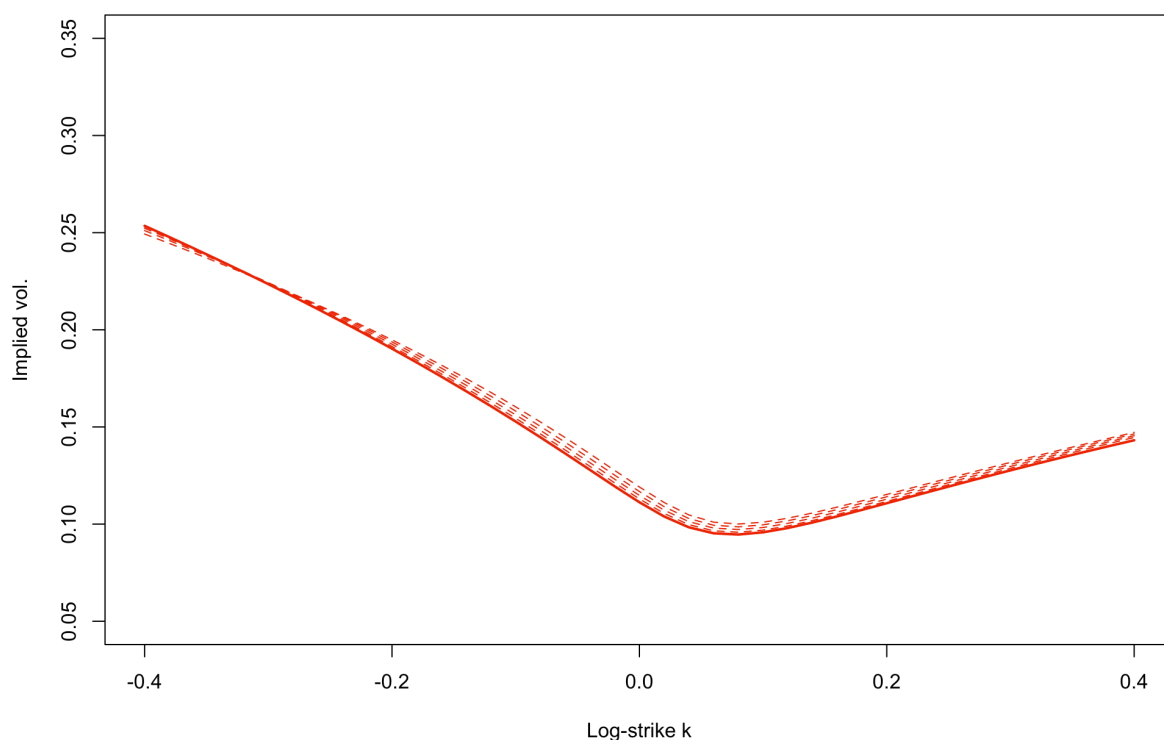


Figure 16: The dotted lines are 1 year smiles with  $H \mapsto H + \{0.1, 0.2, 0.3, 0.4\}$ .

## Sensitivity of the rough Heston 1 week smile to $H$

```
In [51]: subH <- function(incr){
  tmp <- params.rHeston
  tmp$H <- tmp$H+incr
  return(tmp)
}
```

```
vol <- function(params)function(k){
  phi <- phiRoughHestonRational(params, xiCurve, h.approx= h.Pade33, n=100)
  apply(k,function(x){impvol.phi(phi)(x,1/52)}}}
```

```
In [52]: yrange <- c(0.05,.4)
kk <- seq(-.15,.15,.01)
system.time({
  plot(kk,vol(params.rHeston)(kk),col="red2",lwd=2,type="l",
    ylim=yrange,ylab="Implied vol.",xlab="Log-strike k")
  for (incr in seq(0.1,0.4,0.1))
  {
    lines(kk,vol(subH(incr))(kk),lty=2,col="red2")
  }
})
```

```
user system elapsed
2.905 0.029 2.934
```

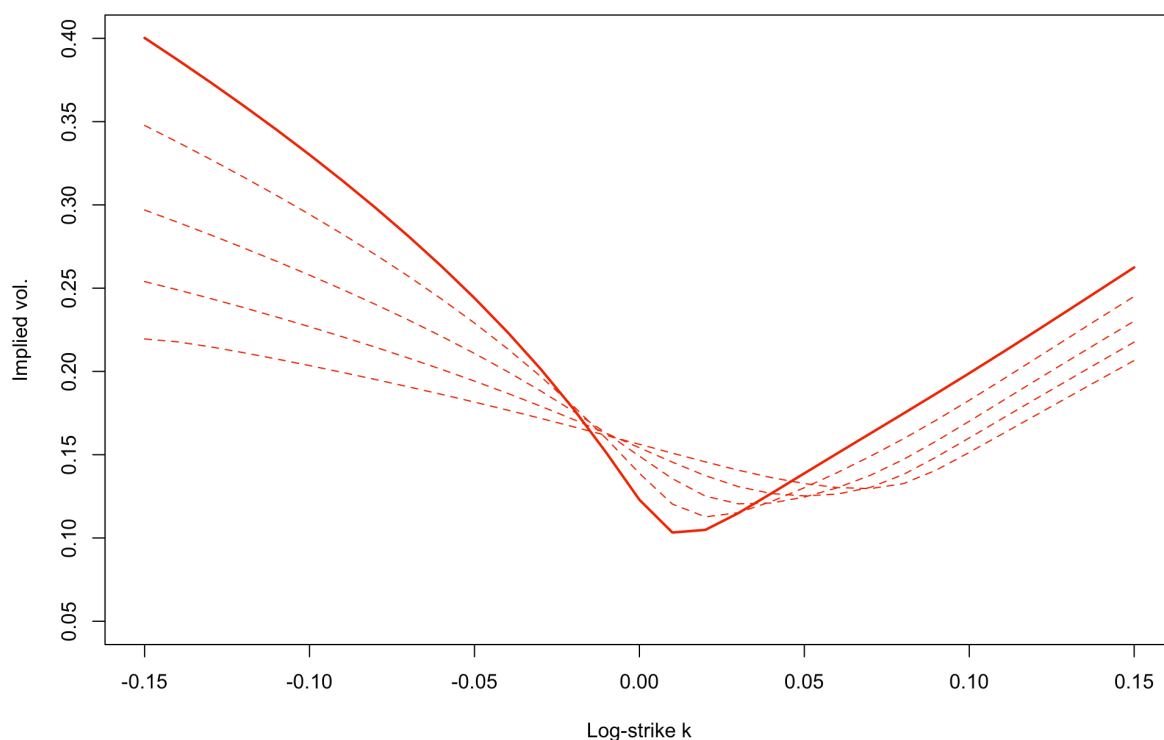


Figure 17: The dotted lines are 1 week smiles with  $H \mapsto H + \{0.1, 0.2, 0.3, 0.4\}$ . The smile flattens as we increase  $H$ .

## Summary of Lecture 3

- We introduced forward variance and affine models.
- We showed how to compute the characteristic function for any affine forward variance model.
- We applied these general results to the classical Heston and rough Heston models.

## References

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